

Homeostatic Synaptic Plasticity of Miniature Excitatory Postsynaptic Currents in Mouse Cortical Cultures Requires Neuronal Rab3A

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Abstract

Following prolonged activity blockade, amplitudes of miniature excitatory postsynaptic currents (mEPSCs) increase, a form of homeostatic plasticity termed “synaptic scaling.” We previously showed that a presynaptic protein, the small GTPase Rab3A, is required for full expression of the increase in miniature endplate current amplitudes following prolonged blockade of action potential activity at the mouse neuromuscular junction *in vivo* (Wang et al., 2011), but it is unknown whether this form of Rab3A-dependent homeostatic plasticity shares any characteristics with central synapses. We show here that synaptic scaling of mEPSCs is impaired in mouse cortical neuron cultures prepared from Rab3A^{-/-} and Rab3A Earlybird mutant mice. To determine if Rab3A is involved in the well-established homeostatic increase in postsynaptic AMPA-type receptors (AMPA-Rs), we performed a series of experiments in which electrophysiological recordings of mEPSCs and confocal imaging of synaptic AMPAR immunofluorescence were assessed within the same cultures. We found that Rab3A is required for the increase in synaptic AMPARs following prolonged activity blockade, but the comparison of mEPSC amplitude and synaptic AMPARs in the same cultures revealed that mEPSC amplitude cannot solely be determined by postsynaptic AMPAR levels. Finally, we demonstrate that Rab3A is acting in neurons because selective loss of Rab3A in astrocytes did not disrupt homeostatic plasticity, whereas selective loss in neurons strongly reduced the homeostatic increase in mEPSC amplitudes. Taken together with the results at the neuromuscular junction, we propose that Rab3A is a presynaptic homeostatic regulator that controls quantal size on both sides of the synapse.

eLife assessment

This study presents **valuable** findings on the role of the small GTPase Rab3A in homeostatic plasticity. While the study demonstrates that Rab3A is required for homeostatic scaling, the evidence supporting the model put forward by the authors is **incomplete**. The work will be of interest to researchers in the field of synaptic transmission and plasticity.

Introduction

One of the most studied phenomena triggered by prolonged activity blockade is the increase in amplitudes of miniature excitatory postsynaptic currents (mEPSCs), first demonstrated in cultures of dissociated cortical neurons (Turrigiano et al., 1998) and spinal cord neurons (O'Brien et al., 1998). This compensatory response is now termed homeostatic synaptic plasticity (Turrigiano and Nelson, 2004; Pozo and Goda, 2010) and has been shown to be dysregulated in several neurodevelopment, psychiatric, and neurodegenerative disorders. One of the initial studies of homeostatic synaptic plasticity demonstrated that AMPAergic mEPSCs were uniformly increased after prolonged activity blockade (Turrigiano et al., 1998), and the authors named this process synaptic scaling. An accompanying increase in AMPA-type glutamate receptors (AMPA receptors) was observed in both of the early studies of homeostatic synaptic plasticity (O'Brien et al., 1998; Turrigiano et al., 1998), and has been confirmed many times (a non-exhaustive list includes (Ju et al., 2004; Thiagarajan et al., 2005; Shepherd et al., 2006; Stellwagen and Malenka, 2006; Hou et al., 2008; Gainey et al., 2009; Soden and Chen, 2010; Correa et al., 2012; Altimimi and Stellwagen, 2013; Letellier et al., 2014; Xu and Pozzo-Miller, 2017).

Homeostatic synaptic plasticity is becoming increasingly implicated in both pathological brain conditions, for example, epilepsy, neuropsychiatric disorders, Huntington's, and alcohol use disorder (Trasande and Ramirez, 2007; Fernandes and Carvalho, 2016; Wang et al., 2017; Lovinger and Abrahao, 2018; Smith-Dijk et al., 2019; Lignani et al., 2020; Suzuki et al., 2021; Kavalali and Monteggia, 2023), and essential normal functions such as sleep (Tononi and Cirelli, 2014; Diering et al., 2017; Torrado Pacheco et al., 2021), so a molecular understanding of the process is an extremely important next step. The homeostatic synaptic plasticity field has identified several proteins required for synaptic scaling of mEPSC amplitudes, the majority of which are involved in the regulation of AMPAR levels (Shepherd et al., 2006; Seeburg and Sheng, 2008; Gao et al., 2010; Anggono et al., 2011; Beique et al., 2011; Diering et al., 2014; Gainey et al., 2015; Tan et al., 2015; Pastuzyn and Shepherd, 2017; Sanderson et al., 2018).

In our previous work studying the increase in miniature endplate currents (mEPC) following prolonged in vivo activity-block of synaptic transmission at the mouse neuromuscular junction (NMJ), we were surprised to find no evidence of changes in acetylcholine receptor (AChR) levels (Wang et al., 2005). This result led us to search for presynaptic molecules that might homeostatically regulate mEPC amplitude, perhaps via the presynaptic quantum. In previous studies in chromaffin cells, we identified the small GTPase Rab3A, a synaptic vesicle protein, as a regulator of synaptic vesicle fusion pore opening (Wang et al., 2008), so we examined whether deletion of Rab3A (Rab3A^{-/-}) might prevent homeostatic upregulation of mEPC amplitude. The results were clear: in the Rab3A^{-/-} mouse, the homeostatic increase in mEPC amplitude was strongly reduced, and was completely abolished in mice expressing a point mutation in Rab3A, the Earlybird mutant (Rab3A^{Ebd/Ebd}) (Wang et al., 2011).

The Rab3A^{-/-} mouse has minimal phenotypic abnormalities, with evoked synaptic transmission and mEPSCs essentially normal in hippocampal slices (Geppert et al., 1994). At the Rab3A^{-/-} NMJ, reductions in evoked transmission were detected, but only under conditions of reduced extracellular calcium (Coleman et al., 2007). The most dramatic effect of loss of Rab3A is the disruption of a presynaptic form of long-term potentiation (LTP) at the mossy fiber-CA3 synapse (Weisskopf et al., 1994; Castillo et al., 1997). To our knowledge, there is no evidence of Rab3A involvement in the expression or trafficking of postsynaptic receptors. Our results at the mammalian NMJ suggest that in addition to its importance in mossy fiber LTP, Rab3A may be required for the homeostatic plasticity of mEPSC amplitude via a presynaptic mechanism.

In the current work, we explored whether the *in vivo* findings at the NMJ might also apply to the more typically studied homeostatic synaptic plasticity of mEPSC amplitude in dissociated cortical neuron cultures. We report that our findings *in vivo* at the NMJ were almost exactly recapitulated in cultures of dissociated cortical neurons: 1. strong reduction of the homeostatic increase in mEPSC amplitude in the absence of Rab3A; 2. complete abolishment of the homeostatic increase in mEPSC amplitude in the presence of the Rab3A Earlybird mutant; and 3. increased mEPSC amplitude in the Rab3A Earlybird mutant prior to activity blockade.

However, in contrast to the unchanged AChR levels at the mammalian NMJ, there was a modest increase in levels of the GluA2-type AMPARs at cortical synapses after activity- blockade, which appeared to be disrupted in cultures prepared from Rab3A^{-/-} mice.

Importantly, when compared within the same cultures, GluA2 receptor levels did not always parallel mEPSC amplitudes. We also determined that Rab3A must be present in neurons, but not astrocytes, for full expression of homeostatic plasticity. The NMJ and cortical culture data taken together strongly suggest neuronal Rab3A is important for both postsynaptic receptor upregulation and a second mechanism affecting mEPSC amplitude, likely the amount of transmitter released by a single vesicle.

Materials and Methods

Animals

Rab3A^{+/-} heterozygous mice were bred and genotyped as previously described (Kapfhamer et al., 2002 [↗](#); Wang et al., 2008 [↗](#)). Rab3A^{Ebd/Ebd} mice were identified in an EU- mutagenesis screen of C57BL/6J mice, and after a cross to C3H/HeJ, were backcrossed for 3 generations to C57BL/6J (Kapfhamer et al., 2002 [↗](#)). Rab3A^{+Ebd} heterozygous mice were bred at Wright State University and genotyped in a two-step procedure: 1. a PCR reaction with RabF1 and Dcaps3R as primers; and 2. a digestion with enzyme Bsp1286I (New England Biolabs) that distinguishes the Earlybird mutant by its different base-pair products. Rab3A^{+/-} mice were backcrossed with Rab3A^{+/+} mice from the Earlybird heterozygous colony for 11 generations in an attempt to establish a single wild type strain, but differences in mEPSC amplitude and adrenal chromaffin cell calcium currents persisted, likely due to genes that are close to the Rab3A site, resulting in two wild type strains: 1. Rab3A^{+/+} from the Rab3A^{+/-} colony, and 2. Rab3A^{+/+} from the Rab3A^{+Ebd} colony.

Primary Culture of Mouse Cortical Neurons

Primary dissociated cultures of mixed neuronal and astrocyte populations were prepared as previously described (Hanes et al., 2020 [↗](#)). Briefly, postnatal day 0-2 (P0-P2) Rab3A^{+/-}, Rab3A^{-/-} or Rab3A^{Ebd/Ebd} neonates were euthanized by rapid decapitation, as approved by the Wright State University Institutional Animal Care and Use Committee, and brains were quickly removed. Each culture was prepared from the cortices harvested from two animals; neonates were not sexed. Cortices were collected in chilled Neurobasal-A media (Gibco) with osmolarity adjusted to 270 mOsm and supplemented with 40 U/ml DNase I (ThermoFisher Scientific). The tissues were digested with papain (Worthington Biochemical) at 20 U/ml at 37°C for 20 minutes followed by trituration with a sterile, fire- polished Pasteur pipette, then filtered through a 100 µm cell strainer, and centrifuged at 1100 rpm for 2 minutes. After discarding the supernatant, the pellet was resuspended in room temperature Neurobasal-A media (270 mOsm), supplemented with 5% fetal bovine serum for astrocyte growth, and 2% B-27 supplement to promote neuronal growth (Gibco), L-glutamine, and gentamicin (ThermoFisher Scientific). Neurons were counted and plated at 0.15 * 10⁶ cells/coverslip onto 12 mm coverslips pre-coated with poly-L-lysine (BioCoat, Corning). The culture media for the first day (0 DIV) was the same as the above Neurobasal-A media supplemented with FBS, B-27, L-glutamine, and gentamicin, and was switched after 24 hours (1 DIV) to media consisting of Neurobasal-A (270 mOsm), 2% B-27, and L-glutamine without

FBS to avoid its toxic effects on neuronal viability and health (Stellwagen and Malenka, 2006 [\[4\]](#)). Half of the media was changed twice weekly and experiments were performed at 13–14 DIV. Two days prior to experiments, tetrodotoxin (TTX) (500 nM; Tocris), a potent Na⁺ channel blocker, was added to some cultures to chronically silence all network activity and induce homeostatic synaptic plasticity mechanisms, while untreated sister cultures served as controls. Cultures prepared from mutant mice were compared with cultures from wild-type mice from their respective colonies. Note that the cultures comprising the Rab3A^{+/+} data here are a subset of the data previously published in Hanes et al., 2020 [\[4\]](#), and therefore the plots in **Figure 1** [\[4\]](#) are not identical to those in the previously published work. This smaller data set was restricted to the time period over which cultures were prepared from Rab3A^{-/-} mice.

Preparation of Astrocyte Feeder Layers

Astrocyte feeder layers were prepared from the cortices of P0–P2 Rab3A^{+/+} or Rab3A^{-/-} mouse pups as described previously (Stellwagen and Malenka, 2006 [\[4\]](#)). Briefly, cortices were dissected and cells were dissociated as described above. Cell suspensions of mixed neuronal and astrocyte populations were plated onto glass coverslips pre-coated with poly-L-lysine in Dulbecco's Modified Eagle Media (ThermoFisher Scientific) supplemented with 5% FBS (to promote astrocyte proliferation and to kill neurons), L-glutamine, and gentamicin, and maintained in an incubator at 37°C, 5% CO₂; cultures were maintained in this manner for up to 1 month to generate purely astrocytic cultures (all neurons typically died off by 7 DIV). Culture media was replaced after 24 hours, and subsequent media changes were made twice weekly, replacing half of the culture media with fresh media. Feeder layers were not used for neuronal seeding until all native neurons were gone and astrocytes approached 100% confluency (visually inspected).

Plating of Neurons on Glial Feeder Layers

Cortical neurons were obtained as described above. The cell pellet obtained was resuspended in Neurobasal-A (osmolarity adjusted to 270 mOsm) containing B27 (2%, to promote neuronal growth), L-glutamine, and 5-fluorodeoxyuridine (FdU, a mitotic inhibitor; Sigma). Addition of FdU was used to prevent astrocyte proliferation and contamination of the feeder layer with new astrocytes, promoting only neuronal growth on the feeder layers (FdU-containing media was used for the maintenance of these cultures and all subsequent media changes). Astrocyte culture media was removed from the feeder layer cultures, and the neuronal cell suspension was plated onto the astrocyte feeder cultures. The culture strategy used to distinguish the relative roles of neuronal and astrocytic Rab3A is outlined in **Figure 9** [\[4\]](#). At 1 DIV, all of the culture media was removed and replaced with fresh Neurobasal-A media containing FdU described above, and half of the media was replaced twice per week for all subsequent media changes. Cultures were maintained in a 37°C, 5% CO₂ incubator for 13–14 DIV.

Whole-Cell Voltage Clamp to Record mEPSCs

At 13–14 DIV, mEPSCs from TTX-treated and untreated sister cultures of Rab3A^{+/+} or Rab3A^{-/-} neurons from the Rab3A^{-/-} colony, or Rab3A^{+/+} or Rab3A^{Ebd/Ebd} neurons from the Rab3A^{+/Ebd} colony, were recorded via whole-cell voltage clamp to assess the role of Rab3A in homeostatic synaptic plasticity. Recordings were taken from pyramidal neurons, which were identified visually by a prominent apical dendrite; images were taken of all cells recorded from. Cells were continuously perfused with a solution consisting of (in mM): NaCl (115), KCl (5), CaCl₂ (2.5), MgCl₂ (1.3), dextrose (23), sucrose (26), HEPES (4.2), pH = 7.2 (Stellwagen and Malenka, 2006 [\[4\]](#)). On the day of recording, the osmolarity of the media from the cultures was measured (normally 285 – 295 mOsm) and the perfusate osmolarity was adjusted to match the culture osmolarity, to protect against osmotic shock to the neurons. To isolate glutamatergic mEPSCs, TTX (500 nM) and picrotoxin (50 μM) were included in the perfusion solution to block action potentials and GABAergic currents, respectively. The NMDA receptor antagonist, APV, was not included in the perfusion solution because all mEPSCs were blocked by CNQX and picrotoxin, demonstrating no

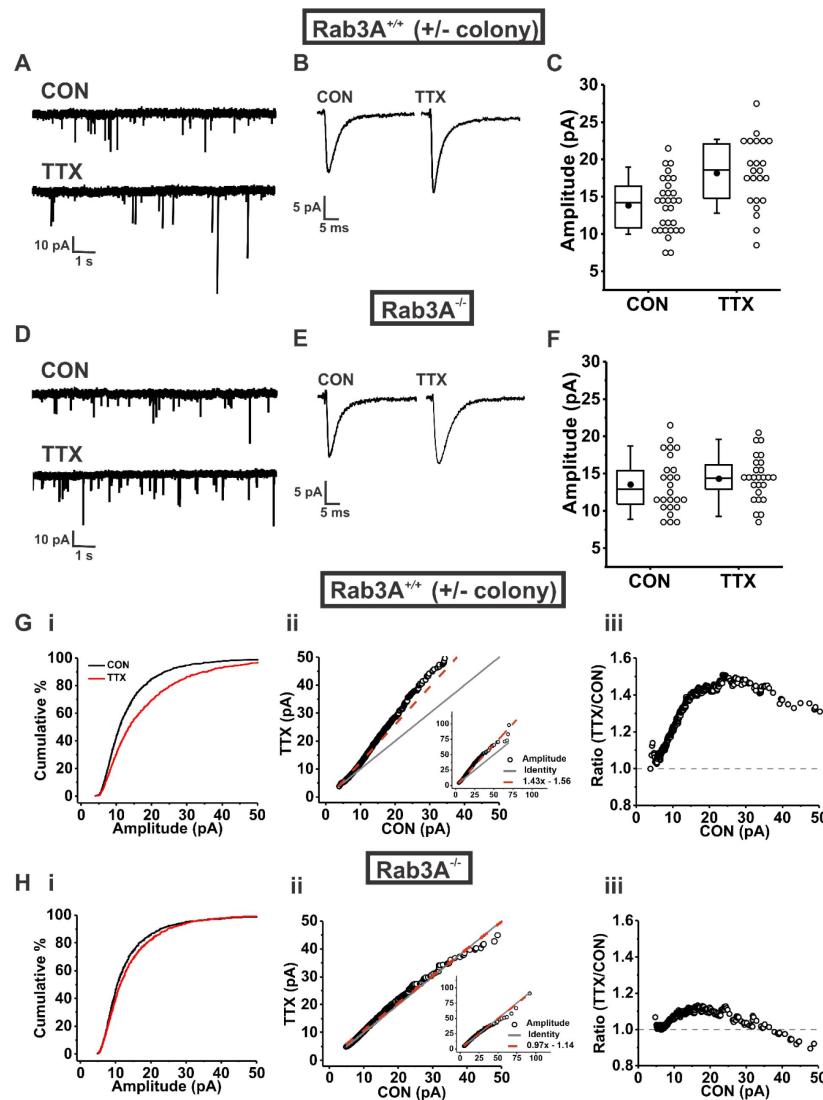


Figure 1.

Block of activity with 48 hours TTX treatment increased mEPSC amplitudes in cortical cultures from wild type mice but not Rab3A^{-/-} mice. A) 10 second current traces recorded at -60 mV in pyramidal cortical neurons from an untreated ("CON") and TTX-treated ("TTX") neuron in cultures prepared from Rab3A^{+/+} mice from the Rab3A^{+/+} colony. B) Average traces for the recordings shown in (A). C) Box and whisker plots for average mEPSC amplitudes from 30 CON and 23 TTX cells (Rab3A^{+/+}, Rab3A^{+/+} colony; mean mEPSC amplitudes, CON, 13.9 ± 0.7 pA; TTX, 18.2 ± 0.9 pA, $p = 4.58 \times 10^{-4}$, Kruskal-Wallis test). D) Current traces recorded in pyramidal cortical neurons from a CON and TTX-treated neuron in cultures prepared from Rab3A^{-/-} mice. E) Average traces for the recordings shown in (D). F) Box and whisker plots for average mEPSC amplitudes from 25 CON and 26 TTX cells, Rab3A^{-/-} data (Rab3A^{-/-}, mean mEPSC amplitudes, CON, 13.6 ± 0.1; TTX, 14.3 ± 0.6, $p = 0.318$, Kruskal-Wallis test). G) Analysis of pooled mEPSC amplitudes for cultures from Rab3A^{+/+} mice of the Rab3A^{+/+} colony (i) Cumulative distribution functions (CDFs) of quantiles of mEPSC amplitudes pooled from CON (black curve) and TTX-treated cultures (red curve) (Kolmogorov Smirnov (KS) test, test statistic (D) = 0.149, $p = 8.53 \times 10^{-13}$). (ii) Quantiles pooled from CON and TTX-treated cells were ranked from smallest to largest, plotted against each other, and fit with linear regression functions. Gray line, identity; red dashed line, linear regression fit. Inset, full range of the data. (iii) The same ranked mEPSC quantiles were paired and the ratio of CON/TTX computed for each pair, then plotted as a function of the CON mEPSC for the pair. Dashed line at ratio = 1.0 (no plasticity). H) Analysis of pooled mEPSCs amplitudes for cultures from Rab3A^{-/-} mice. (i) CDFs of quantiles pooled from CON (black curve) and TTX-treated cultures (red curve) (KS test, D = 0.070, $p = 0.042$). (ii) Quantiles pooled from CON and TTX-treated cells were ranked from smallest to largest, plotted against each other, and fit with a linear regression function. Gray lines, identity; red dashed lines, linear regression fits. Inset: full range of the data. (iii) The same ranked mEPSC quantiles were paired and the ratio of CON/TTX computed for each pair, then plotted as a function of the CON mEPSC for the pair. Dashed line at ratio = 1 (no plasticity).

APV-sensitive mEPSCs were present (data not shown). Patch electrodes (3.5 – 5 M Ω) were filled with an internal solution containing (in mM): K-gluconate (128), NaCl (10), EGTA (1), CaCl₂ (0.132), MgCl₂ (2), HEPES (10), pH = 7.2.

Osmolarity was adjusted to 10 mOsm less than the perfusion solution osmolarity. Neurons were clamped at a voltage of -60 mV using an Axopatch 200B patch-clamp (Axon Instruments), recorded from for 2 – 5 minutes, and data were collected with Clampex 10.0/10.2 (Axon Instruments). The antagonist of Ca²⁺-permeable AMPA receptors (including GluA1 but not GluA2), N-naphthyl acetylspermine (NASPM, 20 μ M; Tocris), was applied during recordings in a subset of experiments. Because NASPM is an open channel blocker, it was applied with a depolarizing high K⁺ solution (25 mM KCl, 95 mM NaCl). Baseline recordings were performed for 2 minutes in our standard perfusate, then were suspended while NASPM + High K⁺ solution was applied for 45 seconds, followed by a NASPM only solution for 5 minutes, after which recording was recommenced for 5 minutes (because we found in pilot experiments that frequency was reduced following NASPM application).

Data Analysis and Statistics of mEPSCs

Miniature excitatory postsynaptic currents were manually selected using Mini Analysis software (Synaptosoft) to identify mEPSCs. The program threshold was set at 3 pA but the smallest mEPSC selected was 4.01 pA. Records were filtered at 2 kHz using a low-pass Butterworth filter prior to selection. For computing mEPSC mean in each experimental condition, individual cell means were pooled across multiple cultures and compared with the non-parametric Kruskal-Wallis test, with n = the number of cells and the overall means presented as $\pm$ SEM. For cumulative probability distribution functions (CDFs) of mEPSC amplitude, 30 quantiles were computed for each cell and pooled across cultures (see (Hanes et al., 2020 [link](#))), and a Kolmogorov-Smirnov test (KS test) was used to test for significant differences, with n = the number of mEPSC quantiles. The rank order plots were created by computing a matched number of quantiles from the two experimental conditions (usually control (CON) and TTX). We used an algorithm to identify the product closest to, but above, 24 quantiles for a data set. For example, if there were 13 CON cells and 12 TTX cells, we computed 24 quantiles for each of the 13 cells and 26 quantiles for each of the 12 cells, for an equal total number of 312 quantiles for both data sets. The quantiles were sorted from smallest to largest, TTX amplitudes plotted vs. CON amplitudes, and the relationship fit with a linear regression function with the intercept term allowed to vary. The ratio plots were created by taking the ratio of TTX/CON and plotting as a function of the CON amplitude quantiles.

Immunocytochemistry, microscopy, and data analysis

Primary cultures of mouse cortical neurons were grown for 13-14 DIV. Antibodies to GluA2 (mouse ab against N-terminal, EMD Millipore) were added directly to live cultures at 1:40 dilution, and incubated at 37 °C in a CO₂ incubator for 45 minutes. Cultures were rinsed 3 times with PBS/5% donkey serum before being fixed with 4% paraformaldehyde. After 3 rinses in PBS/5% donkey serum, cultures were incubated in CY3-labeled donkey-anti-mouse secondary antibodies for 1 hour at room temperature, rinsed in PBS/5% donkey serum, permeabilized with 0.2% saponin, and incubated in chick anti-MAP2 (1:2500, AbCAM) and rabbit anti-VGLUT1 (1:4000, Synaptic Systems) for 1 hour at room temperature in PBS/5% donkey serum. After rinsing with PBS/5% donkey serum, coverslips were incubated with 488-anti chick and CY5-anti rabbit secondary antibodies for 1 hour at room temperature, rinsed, blotted to remove excess liquid, and inverted on a drop of Vectashield (Vector Labs). Coverslips were sealed with nail polish and stored at 4 °C for < 1 week before imaging. All secondary antibodies were from Jackson Immunoresearch and were used at 1:225 dilution.

Coverslips were viewed on a Fluoview FV1000 laser scanning confocal microscope with a 60x oil immersion, 1.35 NA objective. Once a pyramidal neuron was identified, Fluoview 2.1 software was used to zoom in on the primary dendrite (5X) and confocal sections were taken every 0.5 μ m.

Images were analyzed offline with ImagePro 6 (Cybernetics). The composite image was used to locate synaptic sites containing both VGLUT1 and GluA2 immunoreactivity in close apposition to each other and to the primary dendrite or a secondary branch. An area of interest (AOI) was manually drawn around the receptor cluster in the confocal section in which it was the brightest. The AOIs for a dendrite were saved in a single file; the AOI number and the confocal section it was associated with were noted for later retrieval. For quantification, AOIs were loaded, an individual AOI was called up on the appropriate section, and the count/measurement tool used to apply a threshold (400 or 450; identical for CON and TTX coverslips but different for different cultures); pixels within the cluster that were above the threshold were automatically outlined, and size, average intensity, and integral of the outlined region reported. For quantile sampling, only dendrites with a minimum of 6 synaptic sites were included. For creating CDFs, 30 quantiles were computed for each dendrite. For the rank order and ratio plots, we matched the total quantiles for CON and TTX GluA2 receptor characteristics as was done above for mEPSC amplitudes.

Results

We previously reported that mixed cultures of cortical neurons and astrocytes prepared from postnatal day 0-2 mouse pups responded to a block of action potential-mediated activity by a 48 hour TTX treatment with an increase in mEPSC amplitude (Hanes et al., 2020). Here, we asked the question whether cortical cultures prepared from mice lacking the small GTPase Rab3A, or, expressing a point mutation of Rab3A, Rab3A Earlybird, have an altered homeostatic plasticity response to a loss of network activity. To obtain Rab3A^{-/-} and Rab3A^{Ebd/Ebd} homozygotes, we established two mouse colonies of heterozygous breeders with cultures prepared from pups derived from a final breeding pair of homozygotes. Although we backcrossed Rab3A^{+/-} with Rab3A^{+Ebd} for 11 generations, clear differences in mEPSC amplitudes in untreated cultures (see below) and in calcium current amplitudes in adrenal chromaffin cells (unpublished obs.) remained. Therefore, throughout this study we keep the two strains separate and there are two Rab3A^{+/+} or ‘wild type’ phenotypes.

Example current traces of spontaneously occurring mEPSCs recorded from pyramidal neurons in untreated (CON) 13-14 DIV cortical cultures and sister cultures treated with 500 nM TTX for 48 hours prepared from wild type animals in the Rab3A^{+/-} colony are shown in **Figure 1A**. Average mEPSC waveforms from the same recordings are shown in **Figure 1B**. The mean mEPSC amplitudes for 30 control and 23 TTX-treated neurons are displayed in the box and whisker plot in **Figure 1C**; after activity blockade the average mEPSC amplitude increased from 13.9 ± 0.7 pA to 18.2 ± 0.9 pA ($p = 4.58 \times 10^{-4}$, Kruskal-Wallis test). Example current traces of mEPSCs and average mEPSC waveforms are shown in **Figures 1D and E**, respectively, for cortical cultures prepared from Rab3A^{-/-} mice. In contrast to the behavior of neurons in cultures prepared from the wild type strain, the increase in mEPSC amplitudes in Rab3A^{-/-} neurons after activity blockade was dramatically reduced, and the average mEPSC amplitude was not significantly increased, for 25 untreated cells and 26 TTX-treated cells (**Figure 1F**, 13.6 ± 0.1 vs. 14.3 ± 0.6 , $p = 0.318$, Kruskal-Wallis test). To further examine homeostatic plasticity in the presence and absence of Rab3A, we computed 30 quantiles for the mEPSC amplitude distribution of each neuron (Hanes et al., 2020), pooled the quantiles, and plotted the data as cumulative distribution functions (CDFs) for CON and TTX. For mEPSC amplitudes from cultures prepared from Rab3A^{+/+} mice, the difference between CDFs was highly significant (**Figure 1Gi**, test statistic $D = 0.172$, $p = 1.62 \times 10^{-10}$, Kolmogorov Smirnov (KS) test). In contrast, the increase in mEPSC amplitude after activity blockade was dramatically reduced in cortical cultures from Rab3A^{-/-} mice (**Figure 1Hi**). A KS test for the CDFs was significant, but the test statistic much smaller (0.070) and the p value much larger (0.042) than for the wild type strain.

It was originally proposed that loss of activity produced a uniform multiplicative increase in mEPSC amplitude across the entire distribution of mEPSCs (Turrigiano et al., 1998 [\[1\]](#)). This uniform scaling would be expected to preserve original differences in synaptic weights resulting from other forms of plasticity (Turrigiano, 1999 [\[2\]](#); Turrigiano and Nelson, 2004 [\[3\]](#)). In addition, uniform scaling suggested a cell-wide mechanism (in fact, a culture-wide mechanism) to identically modify all synapses. However, we recently showed that in mouse cortical neurons, rat cortical neurons, and mouse hippocampal neurons, scaling is non-uniform, a phenomenon we called “divergent scaling;” the multiplicative factor is smallest (close to 1) for small mEPSC amplitudes and increases to ~1.4 for larger mEPSC amplitudes (Hanes et al., 2020 [\[4\]](#)). In subsequent reviewing of the current literature (Koesters et al., 2022 [\[5\]](#)), we concluded that the majority of previous studies of homeostatic plasticity of mEPSC amplitude following activity blockade also show divergent scaling, most notably the study that originally defined synaptic scaling (Turrigiano et al., 1998 [\[1\]](#)). Divergent scaling is not obvious on the standard plot used to demonstrate synaptic scaling, the rank-order plot (Turrigiano et al., 1998 [\[1\]](#)), but is only apparent once the ratio of TTX mEPSC amplitude/CON mEPSC amplitude is computed and plotted as a function of CON mEPSC amplitude (Hanes et al., 2020 [\[4\]](#)). To compare between uniform and divergent scaling in the presence and absence of Rab3A, we created both rank order plots and ratio plots for Rab3A^{+/+} data. The rank order plot for Rab3A^{+/+} neurons (**Figure 1Gii** [\[6\]](#)) shows that treatment with TTX appears to cause a uniform multiplication of control mEPSC amplitudes based on the linear fit to the data, which had a slope of 1.43 ($R^2 = 0.982$). However, in the ratio plot for mEPSCs in Rab3A^{+/+} cultures, the scaling was clearly divergent with the smallest mEPSC amplitudes having the smallest ratios (at 1) and the ratio increasing with mEPSC amplitude to a maximum of 1.51 (**Figure 1Giii** [\[6\]](#)). In the absence of Rab3A, the rank ordered mEPSC data follow the line of identity and the slope of the linear fit was 0.97 ($R^2 = 0.985$), suggesting complete absence of homeostatic plasticity (**Figure 1Hii** [\[6\]](#)). However, the more sensitive ratio plot reveals a residual divergent scaling to a peak ratio of 1.13 (**Figure 1Hiii** [\[6\]](#)). We previously observed a residual homeostatic effect of activity blockade on mEPSCs at the mouse NMJ in the absence of Rab3A (Wang et al., 2011 [\[7\]](#)) suggesting that there may be compensatory or redundant mechanisms after deletion of Rab3A.

We next examined whether expression of a single point mutant of Rab3A (Rab3A^{Ebd/Ebd}) would abolish synaptic scaling in the mouse cortical cultures as we previously showed for mEPSCs at the mouse NMJ (Wang et al., 2011 [\[7\]](#)). Example current traces of mEPSCs, average mEPSC waveforms, and box and whisker plots are shown in **Figures 2A and B** [\[8\]](#), respectively, for cortical cultures prepared from wild type mice in the Rab3A^{+/Ebd} colony. Treatment with TTX for 48 hours leads to a significant increase in the average mEPSC amplitude of 23 TTX-treated cells when compared to 20 untreated cells (**Figure 2C** [\[8\]](#), CON, 11.0 ± 0.6 pA; TTX, 15.0 ± 1.3 pA, $p = 0.02$, Kruskal-Wallis test). We note here that while the two strains respond very similarly to activity blockade, the mean mEPSC amplitude in untreated cultures was significantly different in the two wild type strains, 13.9 ± 0.7 pA, wild type from Rab3A^{+/+} colony; 11.0 ± 0.6 pA, wild type from Rab3A^{+/Ebd} colony ($p = 0.004$, Kruskal-Wallis test).

We found a complete disruption of homeostatic plasticity in cortical cultures prepared from Rab3A^{Ebd/Ebd} mice, as can be seen in viewing example mEPSC traces and average mEPSC waveforms (**Figures 2D and E** [\[8\]](#), respectively). The lack of TTX effect was confirmed in a comparison of mEPSC amplitude means for 21 untreated and 22 TTX-treated cells (**Figure 2F** [\[8\]](#), CON, 15.1 ± 1.0 pA vs. TTX, 14.6 ± 1.1 pA, $p = 0.81$, Kruskal-Wallis test). For cultures prepared from the wild type strain of the Earlybird heterozygote colony, the CDFs of the pooled quantiles from control and TTX-treated cells were significantly different (**Figure 2Gi** [\[8\]](#), $D = 0.177$, p value = 2.88×10^{-9} , KS test), but for cultures prepared from Rab3A^{Ebd/Ebd} mice, the CDF of mEPSC amplitudes from TTX treated cells was shifted to the *left*, indicating a slight *reduction* after activity blockade, which did not reach statistical significance (**Figure 2Hi** [\[8\]](#); $D = 0.067$, $p = 0.097$, KS test). The rank ordered data indicate that the increased slope after TTX observed in the wildtype data (**Figure 2Gii** [\[8\]](#), slope 1.73, $R^2 = 0.993$) was completely abolished in the Earlybird data (**Figure 2Hii** [\[8\]](#), slope 1.03, $R^2 = 0.989$). Finally, in the wild type data, the ratio of TTX mEPSC amplitude/CON mEPSC

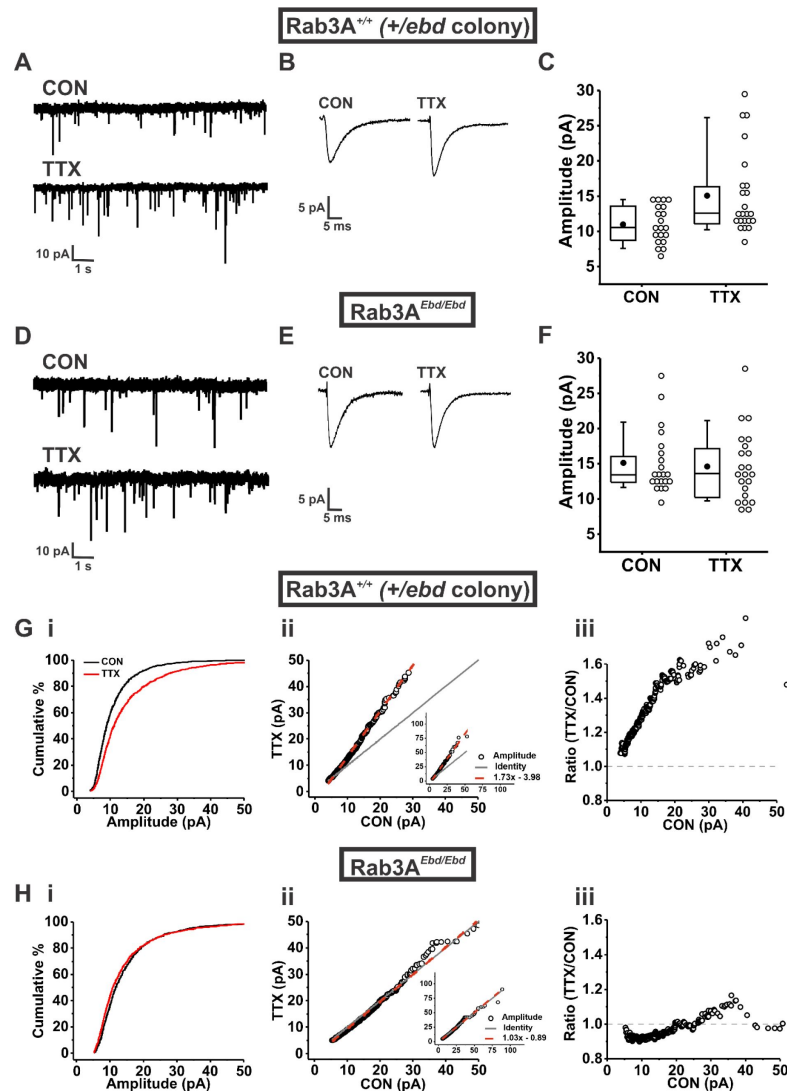


Figure 2.

Block of activity with 48 hours TTX treatment increased mEPSC amplitudes in cortical cultures from wild type mice but not Rab3A^{Ebd/Ebd} mice, which had an increased mEPSC amplitude at baseline. A) 10 second current traces recorded at -60 mV in pyramidal cortical neurons from an untreated ("CON") and TTX-treated ("TTX") neuron in cultures prepared from Rab3A^{+/+} mice from the Rab3A^{+/+}Ebd colony. B) Average traces for the recordings shown in (A). C) Box and whisker plots for average mEPSC amplitudes from 20 CON and 23 TTX cells (Rab3A^{+/+}, Rab3A^{+/+}Ebd colony, mean mEPSCs amplitudes, CON, 11.0 ± 0.6 pA; TTX, 15.0 ± 1.3 pA, p = 0.02, Kruskal-Wallis test). D) Current traces recorded in pyramidal-shaped cortical neurons from a CON and TTX-treated neuron in cultures prepared from Rab3A^{Ebd/Ebd} mice. E) Average traces for the recordings shown in (D). F) Box and whisker plots of average mEPSC amplitudes from 21 CON and 22 TTX cells (Rab3A^{Ebd/Ebd}, mean mEPSC amplitudes, CON, 15.1 ± 1.0 pA; TTX, 14.6 ± 1.1 pA, p = 0.81, Kruskal-Wallis test). G) Analysis of pooled mEPSC amplitudes for cultures from Rab3A^{+/+} mice of the Rab3A^{+/+}Ebd colony. (i) CDFs of quantiles pooled from CON (black curve) and TTX-treated cultures (red curve) (KS test, D = 0.177, p value = 2.88 × 10⁻⁹). (ii) Quantiles pooled from CON and TTX-treated cells were ranked from smallest to largest, plotted against each other, and fit with linear regression function. Gray line, identity; red dashed line, linear regression fit. Inset: full range of the data. (iii) The same ranked mEPSC quantiles were paired and the ratio of CON/TTX computed for each pair, then plotted as a function of the CON mEPSC for the pair. Dashed line at ratio = 1 (no plasticity). H) Analysis of pooled mEPSC amplitudes for cultures from Rab3A^{Ebd/Ebd} mice. (i) CDFs of quantiles pooled from CON (black curve) and TTX-treated cultures (red curve). (KS test, D = 0.067, p value = 0.097 (TTX mEPSCs were slightly reduced)). (ii) Quantiles pooled from CON and TTX-treated cells were ranked from smallest to largest, plotted against each other, and fit with a linear regression function. Gray line, identity; red dashed line, linear regression fit. Inset: full range of the data. (iii) The same ranked mEPSC quantiles were paired and the ratio of CON/TTX computed for each pair, then plotted as a function of the CON mEPSC for the pair. Dashed line at ratio = 1 (no plasticity).

amplitude increases from 1.08 for the smallest amplitudes to over 1.60 for the largest amplitudes (**Figure 2Giii**), whereas for mEPSC amplitudes from Rab3A^{Ebd/Ebd} cultures, the ratio of TTX/CON actually falls below 1 (minimum 0.9, **Figure 2Hiii**).

Our results show that the homeostatic increase of mEPSC amplitude after activity blockade is disrupted in both Rab3A^{-/-} and the Rab3A^{Ebd/Ebd} cortical neurons, strongly supporting a crucial role of functioning Rab3A in the synaptic scaling process. However, it is important to note that the disruption differs for Rab3A^{-/-} and Rab3A^{Ebd/Ebd}. In the Rab3A^{-/-} data set, mEPSCs from untreated cultures were indistinguishable from mEPSCs from Rab3A^{+/+} untreated cultures, demonstrating loss of Rab3A has no impact on basal activity mEPSC amplitudes, but the increase in mEPSC amplitudes after activity blockade was strongly diminished in Rab3A^{-/-} neurons. In the Rab3A^{Ebd/Ebd} data set, mEPSC amplitudes from untreated cultures were significantly larger than those of untreated cultures from wild type mice, as can be seen in **Figure 3** for the CDFs (**Figure 3A**, $D = 0.245$; $p = 1.52 \times 10^{-16}$, KS test) and the box and whisker plot (**Figure 3A** Inset, Rab3A^{+/+}, 11.0 ± 0.7 pA, vs. Rab3A^{Ebd/Ebd}, 15.1 ± 1.0 pA $p = 0.0027$). The linear fit of the rank ordered data of Rab3A^{Ebd/Ebd} vs. wildtype mEPSCs had a slope of 1.63 (**Figure 3B**). In the ratio plot (**Figure 3C**), scaling was more uniform than in any other data set we have examined so far, with values starting at 1.4, declining to 1.2, then rising again to 1.4 for the majority of the data, instead of beginning near 1 and rising to 1.5, as happened for homeostatic plasticity (**Figure 1Giii**). Given this distinct behavior, it cannot be stated that the presence of mutated Rab3A causes the identical effect on mEPSC amplitudes as that of activity blockade. It may be that the much longer time in the presence of the Rab3A Earlybird mutant compared to the 48 hour TTX treatment leads to a stable increase in the smallest mEPSCs.

Alternatively, these are two distinct mechanisms of mEPSC amplitude augmentation that occlude each other. In any case, the increase in mEPSC amplitude in cultures from Rab3A^{Ebd/Ebd} mice is consistent with the increase in mEPSC amplitude we observed at the Rab3A^{Ebd/Ebd} NMJ (Wang et al., 2011).

The small GTPase Rab3A is generally thought to function presynaptically to regulate synaptic vesicle trafficking, possibly in an activity-dependent manner (Castillo et al., 1997; Lonart et al., 1998; Leenders et al., 2001; Schluter et al., 2006; Coleman and Bykhovskaia, 2009; Tian et al., 2012). In contrast, homeostatic plasticity of mEPSC amplitude has been attributed to an increase in postsynaptic receptors on the surface of the dendrite (O'Brien et al., 1998; Turrigiano et al., 1998). Is Rab3A required for the increase in surface AMPA-type glutamate receptors that has been confirmed by multiple studies (see Introduction)? It should be noted that at the NMJ in vivo we could find no evidence for an increase in AChRs after TTX block of the sciatic nerve in vivo (Wang et al., 2005). The type of AMPA receptor that has been shown to be increased, calcium-permeable GluA1-containing AMPA receptor or calcium-impermeable GluA2-containing AMPA receptor, appears to depend on the experimental manipulation. For example, block of activity by APV (NMDA antagonist) combined with TTX (inhibitor of Na channels) induces an increase in mEPSC amplitude in dissociated hippocampal cultures that is completely reversed by acute application of the Ca-permeable receptor-specific inhibitor NASPM (Sutton et al., 2006), indicating the entirety of the homeostatic effect is due to those receptors, likely GluA1. In contrast, the increase in mEPSC amplitude in mouse hippocampal slice cultures induced by TTX alone was not affected by another GluA1-specific inhibitor, philanthotoxin, suggesting mediation by either Ca-impermeable receptors or a presynaptic effect on quantal size (Soden and Chen, 2010; see also, (Dubes et al., 2022)). Because we used TTX alone to block network activity, we expected that NASPM would not reverse the TTX-induced increase in mEPSC amplitude in our mouse cortical cultures, and we found that this was indeed the case. **Figure 4A** shows that the TTX-induced increase in mean mEPSC amplitude was nearly identical in a set of 11 CON and 11 TTX-treated cells before and after NASPM treatment (before NASPM, CON 12.9 ± 3.5 pA; TTX, 17.5 ± 3.1 pA, $p = 0.009$; after NASPM, CON 11.9 ± 2.6 pA; TTX 16.1 ± 3.5 pA, $p = 0.006$, Kruskal-Wallis test). Application of NASPM caused a modest decrease in mEPSC amplitude in both untreated and TTX-

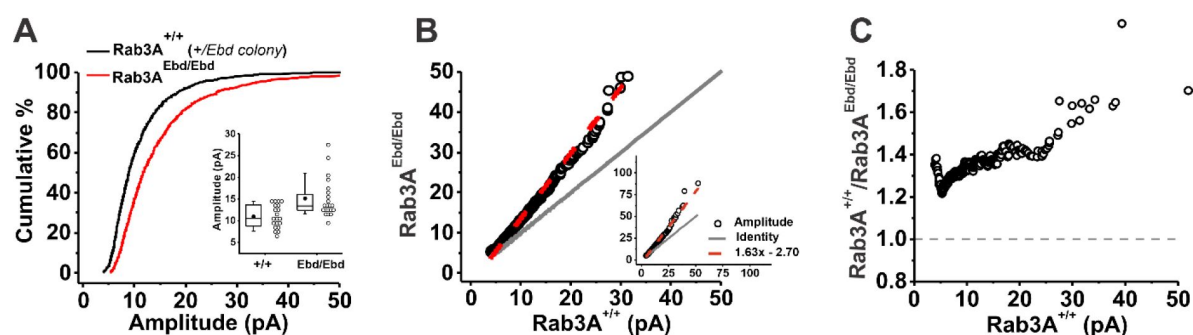


Figure 3.

The increase in mEPSC amplitudes in neurons from cultures prepared from *Rab3A*^{Ebd/Ebd} mice, relative to those from *Rab3A*^{+/+} mice in the same strain, showed less divergent scaling than that of the activity-block induced increase in mEPSC amplitudes typically observed in cultures from *Rab3A*^{+/+} mice. A) Cumulative distribution function of quantiles pooled from *Rab3A*^{+/+} (black curve) and *Rab3A*^{Ebd/Ebd} cultures (red curve), $D = 0.245$; $p = 1.52 \times 10^{-16}$, KS test. Inset: box and whisker plot of average mEPSC amplitudes from 20 *Rab3A*^{+/+} and 21 *Rab3A*^{Ebd/Ebd} cells, mean mEPSC amplitudes, *Rab3A*^{+/+}, 11.0 ± 0.7 pA; *Rab3A*^{Ebd/Ebd}, 15.1 ± 1.0 pA, $p = 0.0027$, Kruskal-Wallis test. B) Quantiles pooled from *Rab3A*^{+/+} and *Rab3A*^{Ebd/Ebd} cells were ranked from smallest to largest, plotted against each other, and fit with linear regression functions. Gray line, identity; red dashed line, linear regression fit. C) The same ranked mEPSC quantiles were paired and the ratio of *Rab3A*^{+/+}/*Rab3A*^{Ebd/Ebd} computed for each pair, then plotted as a function of the *Rab3A*^{+/+} mEPSC for the pair. Dashed line at ratio = 1.0 (no difference).

treated cultures (**Figure 4B**, CON, before NASPM, 12.9 ± 3.5 pA; after, 11.9 ± 2.6 pA, $p = 0.08$; TTX, before NASPM, 17.5 ± 3.1 pA; after, 16.1 ± 3.5 pA, $p = 0.08$, paired t test), and also decreased mEPSC frequency (**Figure 4C**, CON, before NASPM, 1.84 sec^{-1} ; after NASPM, 1.56 sec^{-1} ; $p = 0.003$, paired t-test; TTX, before NASPM, $4.40 \pm 3.51 \text{ mEPSCs sec}^{-1}$; after NASPM, $2.68 \pm 2.25 \text{ mEPSCs sec}^{-1}$, $p = 0.02$, paired t-test), indicating that effective concentrations of NASPM were reached and mEPSCs due to AMPA receptors composed entirely of Ca-permeable subunits were abolished.

Having established that GluA1 receptors are not contributing to the homeostatic increase in mEPSC amplitude, we turned to immunohistochemistry and confocal imaging to assess whether GluA2 receptor expression was increased in our wild type mouse cortical cultures following 48 hour treatment with TTX. Since mEPSCs necessarily report synaptic levels of receptors, we used VGLUT1-positivity to identify synapses on pyramidal primary apical dendrites labeled with MAP-2 immunofluorescence. **Figure 5** shows 6 pairs of VGLUT1- and GluA2-immunofluorescent clusters (white frames) along each of two primary dendrites, one in an untreated cortical culture (CON, left), the other in a culture treated with TTX for 48 hours (TTX, right), both from Rab3A^{+/+} mice. A dendrite typically required ~10 confocal sections to be fully captured, and the total number of synaptic pairs for all the sections imaged on a dendrite was usually < 20, so this is an atypically high number of pairs within a single section; these particular dendrites and sections were selected for illustration purposes. In addition to the synaptic pairs, we observed many GluA2-immunoreactive clusters not associated with VGLUT1 immunoreactivity; those along the dendrites may be extrasynaptic receptors, and those outside the dendrites may be on astrocytes (Fan et al., 1999). There are also GluA2 immunoreactive clusters that are close to VGLUT1 immunoreactivity but are not located along any apparent MAP2-positive neurite, suggesting axon-axonal contacts, although VGLUT1 has also been detected in astrocytes (Ormel et al., 2012). Only sites that contained both VGLUT1 and GluA2 immunoreactivity close to the primary MAP2-positive dendrite or a secondary branch were selected for analyses.

Variability in the magnitude of the homeostatic response from culture to culture is averaged out in physiological experiments by the pooling of data from many cells across many cultures. To reduce the necessity for many cultures, we chose to pair experiments in the same cultures by recording mEPSCs from one set of coverslips, and processing another set of coverslips from the same culture for immunohistochemistry. We completed this matched paradigm of physiology and immunohistochemistry, which to our knowledge has never been done before, for 3 cultures prepared from Rab3A^{+/+} mice and 3 cultures prepared from Rab3A^{-/-} mice. We present the results for mEPSC data and imaging data pooled across the 3 experiments in **Figure 6** (Rab3A^{+/+}), **Figure 7** (Rab3A^{-/-}), and **Table 1**. Levels of GluA2 immunoreactivity at synaptic sites were quantified by the size of the GluA2-positive receptor cluster and the average intensity value of the receptor cluster. We analyzed the imaging data the same way we analyzed mEPSC data, sampling 30 quantiles from each dendrite's data set, and calculating the means, pooling quantiles across dendrites to create CDFs, and sorting quantiles from smallest to largest to produce rank order and ratio plots.

In the data pooled from the 3 matched experiments, neurons from Rab3A^{+/+} cultures showed a significant increase in mean mEPSC amplitudes following activity blockade (**Figure 6A** inset, CON, 13.7 ± 4.5 pA, $n = 23$; TTX, 16.4 ± 4.3 pA, $n = 24$; $p = 0.016$, Kruskal-Wallis test); a significant shift of the TTX CDF to larger mEPSC amplitude values (**Figure 6A**, $D = 0.162$, $p = 1.42 \times 10^{-8}$, KS test); a slope of 1.11 in the rank order plot (**Figure 6B**, $R^2 = 0.98$); and a mean ratio of the 50th to 75th percentile of 1.24 in the ratio plot (**Figure 6C**). These data indicate that the homeostatic response averaged across the 3 cultures was very similar to, but slightly smaller than, that of the previous data set presented in **Figure 1**. The means for size and intensity of GluA2 receptor clusters, while showing trends to higher values after activity blockade, were not significantly different (**Figure 6D** inset, size, CON, $0.97 \pm 0.38 \mu\text{m}^2$; TTX, $1.15 \pm 0.59 \mu\text{m}^2$, $p = 0.44$; **Figure 6G** inset, intensity, CON, 673 ± 90 , TTX, 687 ± 72 , $p = 0.25$, Kruskal-Wallis test; **Table 1**). However, when comparing the CDFs, where the sample size is much greater (equal to usually close to 30

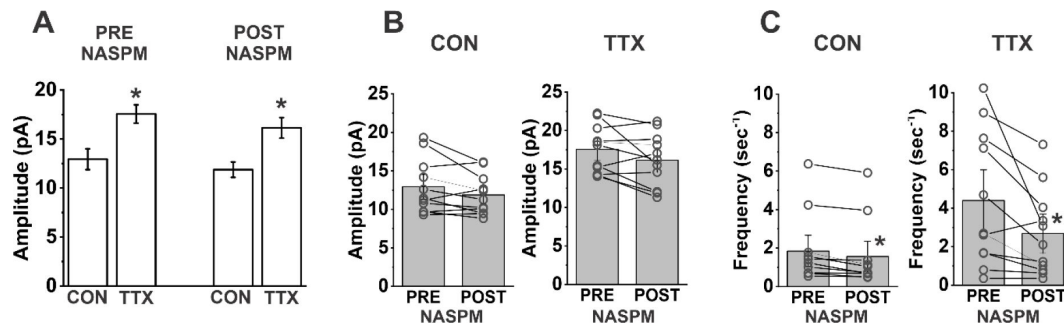


Figure 4.

Ca²⁺-permeable, NASPM-sensitive glutamate receptors did not mediate the TTX-induced increase in mEPSC amplitude in cultures of dissociated mouse cortical neurons. A) mEPSC amplitudes for 11 mouse cortical neurons from untreated cultures ("CON") and 11 neurons from TTX-treated cultures ("TTX") were recorded before and after application of 20 μ M NASPM. mEPSC amplitudes were still significantly larger in TTX-treated cultures compared to untreated control cultures after NASPM treatment (TTX effect pre-NASPM, $p = 0.009$; TTX effect post-NASPM, $p = 0.006$, Kruskal-Wallis non-parametric test). B) Acute application of NASPM had a modest effect on mEPSC amplitude in cells from CON or TTX cultures that did not reach statistical significance (CON, pre-NASPM, 12.9 ± 1.1 pA, post-NASPM, 11.9 ± 0.8 pA, $p = 0.08$; TTX, pre-NASPM, 17.5 ± 0.9 pA, post-NASPM, 16.1 ± 1.0 pA, $p = 0.08$; student's paired t-test). C) Acute application of NASPM consistently reduced mEPSC frequency in cells from untreated and cells from TTX-treated cultures (CON, pre-NASPM, 1.84 ± 1.82 mEPSCs sec⁻¹, post-NASPM, 1.56 ± 1.74 mEPSCs sec⁻¹; $p = 0.003$; TTX, pre-NASPM, 4.40 ± 3.51 mEPSCs sec⁻¹, post-NASPM, 2.68 ± 2.25 mEPSCs sec⁻¹, $p = 0.02$; student's paired t-test).

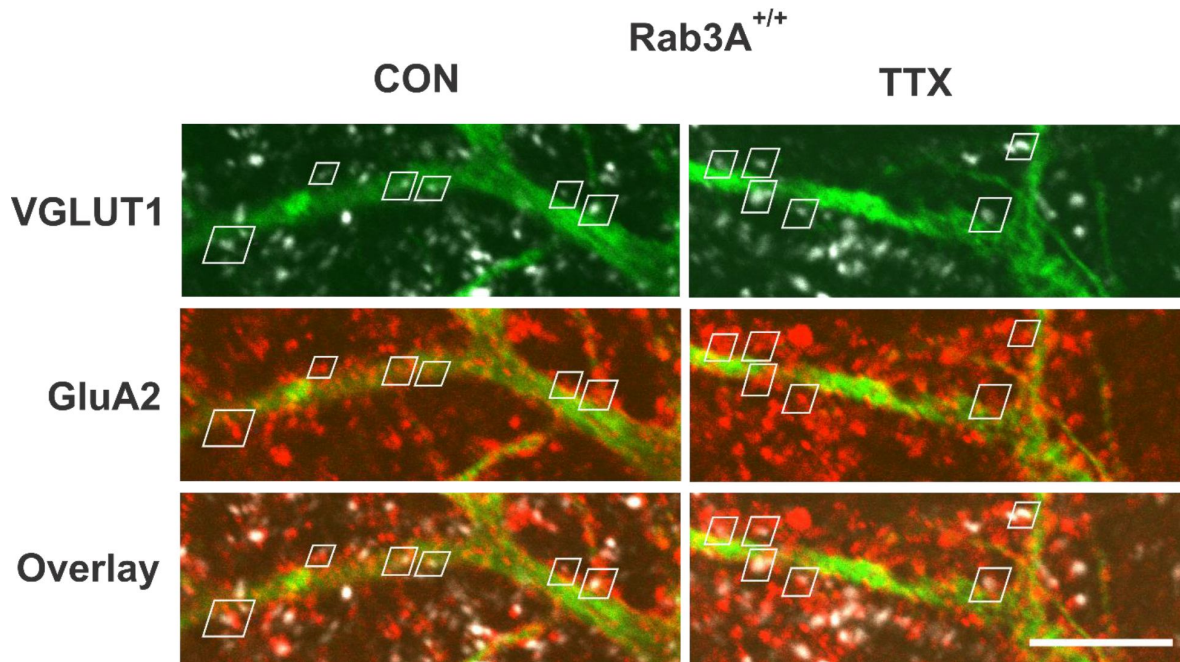


Figure 5.

Confocal images of GluA2 and VGLUT1 immunofluorescence on MAP2-positive primary dendrites of pyramidal-shaped neurons in high density dissociated cortical cultures prepared from Rab3A^{+/+} mice. Shown are 5X confocal sections of two primary dendrites, one from an untreated culture (CON) and the other from a culture treated with TTX (TTX). Postsynaptic GluA2-positive receptor clusters were identified by their proximity to VGLUT1 immunoreactive presynaptic terminals within the same confocal section. GluA2-VGLUT1 synaptic pairs, which were of maximal brightness in these confocal slices, are indicated by white trapezoids. It can be seen that substantial numbers of GluA2-positive clusters located on the dendrites are unopposed by VGLUT1-positive terminals, and that some GluA2-positive and VGLUT1-positive clusters are found away from any MAP2 positivity, suggesting localization to axons or astrocytes. Scale bar = 10 μ m.

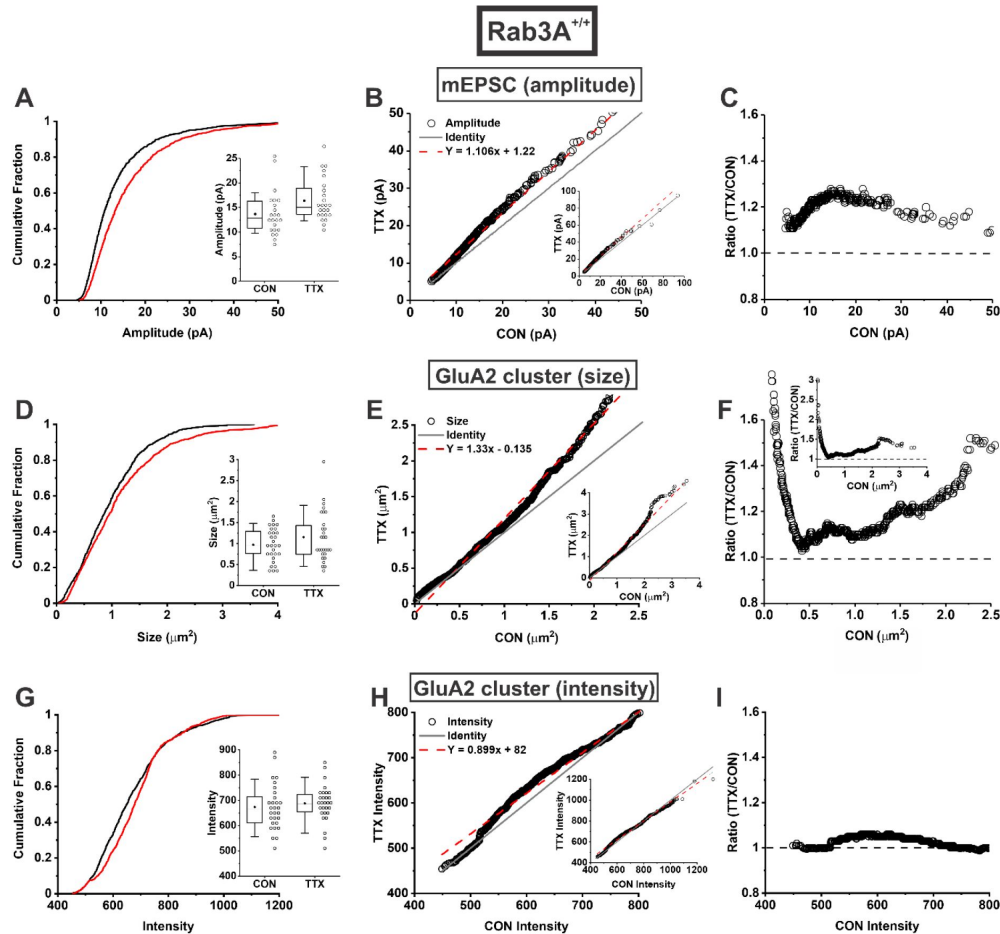


Figure 6.

GluA2 receptor cluster size and average intensity increased following activity blockade in parallel with mEPSC amplitudes recorded in the same cortical cultures prepared from Rab3A^{+/+} mice, but the effects on receptor levels were less robust and differed from mEPSC amplitudes in characteristics of scaling. A) Cumulative distribution functions (CDFs) of mEPSC amplitudes recorded from untreated ("CON") and TTX-treated cells ("TTX") pooled from 3 cultures were significantly different (KS test, $D = 0.162$, $p = 1.42 \times 10^{-8}$, CON, $n = 690$, TTX, $n = 720$). Inset, mean mEPSC amplitudes for 23 CON cells and 24 TTX cells were significantly different (CON, 13.7 ± 4.5 pA; TTX, 16.4 ± 4.3 pA, $p = 0.016$, Kruskal-Wallis test). B) 24 mEPSC amplitude quantiles from each of 23 untreated cells were sorted from smallest to largest and plotted vs. 23 mEPSC amplitude quantiles from each of 24 TTX-treated cells, and the resulting relationship fit with a linear regression equation, $Y = 1.106x + 1.22$ ($R^2 = 0.984$). Inset, full range of data used for fit. C) Sorted data from (B) were used to calculate the ratio of TTX mEPSC amplitude/CON mEPSC amplitude and the ratios plotted vs. CON mEPSC amplitude. D) CDFs of size of GluA2 receptor clusters imaged from primary dendrites in the same untreated cultures and TTX-treated cultures as in (A) – (C) were significantly different (KS test, $D = 0.089$, $p = 0.002$, CON, $n = 870$, TTX, $n = 870$). Inset, mean GluA2 cluster size for 29 CON dendrites and 29 TTX dendrites were not significantly different (CON, $0.97 \pm 0.38 \mu\text{m}^2$; TTX, $1.15 \pm 0.59 \mu\text{m}^2$, $p = 0.44$, Kruskal-Wallis test). E) 29 GluA2 cluster size quantiles from each of 29 untreated dendrites were sorted from smallest to largest and plotted vs. 29 cluster size quantiles from each of 29 TTX-treated dendrites, and the resulting relationship fit with a linear regression equation, $Y = 1.328x - 0.136$ ($R^2 = 0.98$). Inset, full range of data used for fit. F) Sorted data from (E) were used to calculate the ratio of TTX GluA2 cluster size/CON GluA2 cluster size and the ratios plotted vs. CON GluA2 cluster size. G) CDFs of intensity of GluA2 receptor clusters imaged from primary dendrites in untreated cultures and TTX-treated cultures were significantly different (KS test, $D = 0.120$, $p = 6.73 \times 10^{-6}$, CON, $n = 870$, TTX, $n = 870$). Inset, mean GluA2 cluster intensity for 29 CON dendrites and 29 TTX dendrites were not significantly different (CON, 673 ± 90 a.u.; TTX, 687 ± 72 a.u., $p = 0.25$, Kruskal-Wallis test). H) 29 GluA2 cluster intensity quantiles from 29 untreated dendrites were sorted from smallest to largest and plotted vs. 29 cluster intensity quantiles from 29 TTX-treated dendrites, and the resulting relationship fit with a linear regression equation, $Y = 0.899x + 82$ ($R^2 = 0.987$). Inset, full range of data used for fit. I) Sorted data from (H) were used to calculate the ratio of TTX cluster intensity/CON cluster intensity and plotted vs. CON cluster intensity.

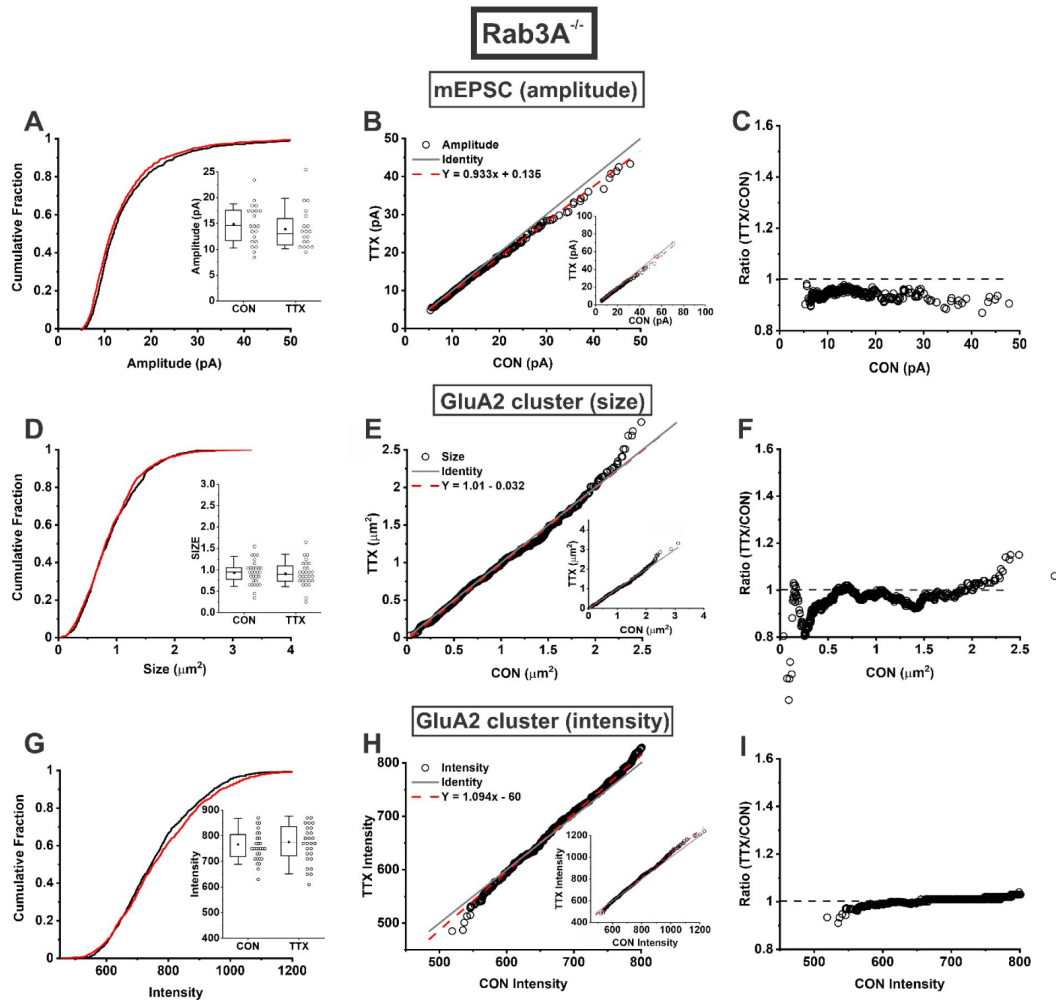


Figure 7.

GluA2 receptor cluster size, average intensity, and mEPSC amplitudes recorded in the same cortical cultures prepared from Rab3A^{-/-} mice were not increased following activity blockade. A) Cumulative distribution functions (CDFs) of mEPSC amplitudes recorded from untreated (“CON”) and TTX-treated cells (“TTX”) pooled from 3 Rab3A^{-/-} cultures were not significantly different (KS test, $D = 0.072$, $p = 0.085$, CON, $n = 630$, TTX, $n = 570$). *Inset*, mean mEPSC amplitudes for 21 CON cells and 19 TTX cells were not significantly different (CON, 14.9 ± 3.8 pA; TTX, 14.0 ± 4.0 pA, $p = 0.34$, Kruskal-Wallis test). B) 19 mEPSC quantiles from each of 21 untreated cells were sorted from smallest to largest and plotted vs. 21 mEPSC quantiles from each of 19 TTX-treated cells, and the resulting relationship fit with a linear regression equation, $Y = 0.93x + 0.13$ ($R^2 = 0.997$). *Inset*, full range of data used for fit. C) Sorted data from (B) were used to calculate the ratio of TTX mEPSC amplitude/CON mEPSC amplitude and plotted vs. CON mEPSC amplitude. D) CDFs of GluA2 receptor cluster size imaged from primary dendrites in the same untreated cultures and TTX-treated cultures as in (A) – (C) were not significantly different (KS test, $D = 0.045$, $p = 0.31$, CON, $n = 900$, TTX, $n = 870$). *Inset*, mean GluA2 cluster size for 30 CON dendrites and 29 TTX dendrites were not significantly different (CON, 0.93 ± 0.27 μm²; TTX, 0.91 ± 0.28 μm², $p = 0.74$, Kruskal-Wallis test). E) 29 GluA2 cluster size quantiles from each of 30 untreated dendrites were sorted from smallest to largest and plotted vs. 30 cluster size quantiles from each of 29 TTX-treated dendrites, and the resulting relationship fit with a linear regression equation, $Y = 1.01x - 0.03$ ($R^2 = 0.993$). *Inset*, full range of data used for fit. F) Sorted data from (E) were used to calculate the ratio of TTX GluA2 cluster size/CON GluA2 cluster size and the ratios plotted vs. CON GluA2 cluster size. G) CDFs of GluA2 receptor cluster average intensity imaged from primary dendrites in untreated cultures and TTX-treated cultures were not significantly different (KS test, $D = 0.060$, $p = 0.08$, CON, $n = 900$, TTX, $n = 870$). *Inset*, mean GluA2 cluster intensity for 30 CON dendrites and 29 TTX dendrites were not significantly different (CON, 766 ± 68 a.u.; TTX, 776 ± 79 a.u., $p = 0.47$, Kruskal-Wallis test). H) 29 GluA2 cluster intensity quantiles from 30 untreated dendrites were sorted from smallest to largest and plotted vs. 30 cluster intensity quantiles from 29 TTX-treated dendrites, and the resulting relationship fit with a linear regression equation, $Y = 1.09x - 60$ ($R^2 = 0.993$). *Inset*, full range of data used for fit. I) Sorted data from (H) were used to calculate the ratio of TTX cluster intensity/CON cluster intensity and the ratios plotted vs. CON cluster intensity.

	Rab3A^{+/+}		Rab3A^{-/-}	
mEPSC amplitude (pA)	CON	TTX	CON	TTX
mean	13.7 ± 4.5 (n=23)	16.4 ± 4.3 (n=24)	14.9 ± 3.8 (n=21)	14.0 ± 4.0 (n=19)
KW p value	p = 0.016		p = 0.34	
KS comparison of CDF	N CON (n=690), N TTX (n=720) D = 0.162 p = 1.42 * 10 ⁻⁸		N CON (n=630), N TTX (n=570) D = 0.072 p = 0.085	
Rank order linear regression (slope, intercept, R ²)	Slope, 1.11 Intercept, 1.22 R ² , 0.984		Slope, 0.93 Intercept, 0.13 R ² , 0.997	
Mean ratio TTX/CON 50-75 th quantile	1.24		0.957	
GluA2 receptor cluster size (μm²)	CON	TTX	CON	TTX
mean	0.97 ± 0.38 (n=29)	1.15 ± 0.59 (n=29)	0.93 ± 0.27 (n=30)	0.91 ± 0.28 (n=29)
KW p value	p = 0.44		p = 0.74	
KS comparison of CDF	N CON (n=870), N TTX (n=870) D = 0.089 p = 0.002		N CON (n=900), N TTX (n=870) D = 0.045 p = 0.31	
Rank order linear regression (slope, intercept, R ²)	Slope, 1.33 Intercept, -0.13 R ² , 0.98		Slope, 1.01 Intercept, -0.03 R ² , 0.993	
Mean ratio TTX/CON 50-75 th quantile	1.11		0.976	
GluA2 receptor cluster intensity (arb. units)	CON	TTX	CON	TTX
mean	673 ± 90 (n=29)	687 ± 72 (n=29)	766 ± 68 (n=30)	776 ± 79 (n=29)
KW p value	p = 0.25		p = 0.47	
KS comparison of CDF	N CON (n=870), N TTX (n=870) D = 0.120 p = 6.73 * 10 ⁻⁶		N CON (n=900), N TTX (n=870) D = 0.060 p = 0.08	
Rank order linear regression (slope, intercept, R ²)	Slope, 0.90 Intercept, 82 R ² , 0.987		Slope, 1.09 Intercept, -60 R ² , 0.993	
Mean ratio TTX/CON 50-75 th quantile	1.02		1.02	

Table 1.

Comparison of effect of activity blockade with TTX on mEPSC amplitude and GluA2 receptor cluster characteristics. Voltage clamp recordings of mEPSCs and confocal imaging of GluA2 receptor immunofluorescence at VGLU1-positive synapses were performed on coverslips from the same 3 Rab3A^{+/+} cultures (left) and 3 Rab3A^{-/-} cultures (right). p values are not corrected for multiple comparisons.

quantiles * number of cells), the shifts to larger values for the TTX CDF did reach statistical significance for GluA2 receptor cluster size (**Figure 6D**, $D = 0.089$, $p = 0.002$, KS test) and intensity (**Figure 6G**, $D = 0.120$, $p = 6.73 \times 10^{-6}$, KS test). In the rank order plot for GluA2 receptor cluster sizes (**Figure 6E**), the slope was actually larger than that for mEPSC amplitude (1.33), but for intensity (**Figure 6H**), the slope value was below 1 (0.90). The weaker effects on synaptic GluA2 receptor levels relative to mEPSC amplitude suggest the possibility that GluA2 receptors are not the sole determining factor in the increased mEPSC amplitude following activity blockade. In addition, the plot of the ratio of TTX GluA2 receptor cluster size/control size as a function of ranked control values showed a surprising deviation from the divergent scaling we have observed for mEPSC amplitude (compare **Figure 6C** to **Figure 6F**). The smallest GluA2 receptor clusters showed a TTX/CON ratio as high as 3, but the mean ratio across the 50th to 75th percentile settled down to 1.11 within < 20 samples. This finding of larger ratios for the smallest GluA2 receptor clusters is supported by a previous study that followed the same fluorescently labeled postsynaptic sites over time following activity blockade. Wang and colleagues found that the biggest increases after activity blockade occurred at the smallest/dimmest synaptic sites (Wang et al., 2019). A similar inverse relationship was reported for changes in individual fluorescently labeled postsynaptic sites followed over time under normal activity conditions (Minerbi et al., 2009; Statman et al., 2014). The mismatch in ratio values between the smallest mEPSC amplitudes and smallest GluA2 receptor cluster sizes may be due to the inability of physiological assays to detect the mEPSCs coming from the smallest synapses.

We proceeded to determine the homeostatic responses of mEPSC amplitude and GluA2 receptor levels in 3 cultures prepared from Rab3A^{-/-} mice. As shown for the data set in **Figure 1**, mean mEPSC amplitude was not increased following activity blockade in the data pooled from this new set of 3 Rab3A^{-/-} cultures (**Figure 7A** inset, CON 14.9 ± 3.8 pA, $n = 21$; TTX, 14.0 ± 4.0 pA, $n = 19$; $p = 0.34$, Kruskal-Wallis test). Notably, the TTX CDF was shifted to *smaller* mEPSC amplitude values (**Figure 7A**, $D = 0.072$, $p = 0.085$, KS test), the slope of the rank order plot was below 1 (**Figure 7B**, slope 0.93, $R^2 = 0.997$), and the mean ratio across the 50th to 75th percentile was below 1 (**Figure 7C**, 0.96). Thus, the disruption of homeostatic plasticity of mEPSC amplitude shown with the previous data sets was recapitulated in this data set. For immunohistochemistry results from the same Rab3A^{-/-} cultures, we found that mean GluA2 receptor cluster size was unchanged following activity blockade (**Figure 7D** inset, CON, $0.93 \pm 0.27 \mu\text{m}^2$, TTX, $0.91 \pm 0.28 \mu\text{m}^2$, $p = 0.74$, Kruskal-Wallis test), the TTX CDF was not significantly shifted (**Figure 7D**, $D = 0.045$, $p = 0.31$, KS test), the slope on the rank order plot was 1.01 (**Figure 7E**), and the ratio hovered at or below 1 except for a small group of ratios at the high end of the control values (**Figure 7F**). For intensity, mean values were not increased after activity blockade (**Figure 7G** inset, CON, 766 ± 68 , TTX, 776 ± 79 , $p = 0.47$, Kruskal-Wallis test), the CDFs were not significantly different (**Figure 7G**, $D = 0.060$, $p = 0.080$), the slope on the rank order plot was 1.09 (**Figure 7H**), and the mean ratio value across the 50th to 75th percentile was 1.02 (**Figure 7I**). Taken together, these results indicate that the modest increases in GluA2 receptor cluster size and intensity following activity blockade observed in wild-type cultures do not occur in the absence of Rab3A.

As noted above, the magnitude of the homeostatic effect on mEPSC amplitude appeared to be more robust than that on receptor levels in cultures from Rab3A^{+/-} mice. Further evidence that there is not a one-to-one correspondence in the homeostatic response of mEPSC amplitudes and GluA2 receptor levels was apparent when the individual experiments' data were compared. The electrophysiology experiments were technically difficult because in order to evaluate whether a particular culture displayed homeostatic plasticity of mEPSC amplitude, we set a minimum requirement of recording from 6 cells per condition, meaning a successful experiment required at least 12 (6 CON and 6 TTX) usable recordings (i.e. low holding current, stable baseline, low noise, etc.) in one day. Ultimately, our sample size ranged from 6 to 10 cells per condition per culture. **Figures 8A-C** show the CDFs and ratio plots for 3 individual Rab3A^{+/-} cultures. Rank order plots are not included here because the ratio plot is more sensitive than the rank order plot when determining effect magnitude. We found that the mEPSC amplitude effects differed from the

GluA2 cluster size effects at both the broad level (increases vs decreases), and in the specific ways in which the effect was non-uniform. Broadly, although two of the three Rab3A^{+/+} experiments show a homeostatic increase in the TTX CDF and a ratio of TTX/CON > 1 for both mEPSC amplitudes (**Figures 8Ai**,iii and **8Bi**,iii) and synaptic GluA2 receptor cluster sizes (**Figures 8Aii**,iv and **8Bii**,iv), Culture #3 shows an increase in mEPSC amplitude but a *decrease* in the GluA2 receptor cluster size, based on both the shifts in CDFs (mEPSC, **Figure 8Ci**, vs GluA2 receptor cluster size, **Figure 8Cii**) and the majority of ratio values (mEPSC, **Figure 8Ciii**, vs. GluA2 receptor cluster size, **Figure 8Civ**). Similarly, we found an obvious mismatch between the mEPSC amplitude data and the GluA2 cluster size data for one of the three Rab3A^{-/-} experiments. For Culture #3, the mEPSC TTX CDF shows a shift to the left (**Figure 8Fi**), and the ratios were below 1 for control mEPSC amplitudes < 20 pA and above 1 for control mEPSC amplitudes > 20 pA (**Figure 8Fiii**). In the same culture, GluA2 receptor cluster size CDF shows a shift to the right after activity blockade (**Figure 8Fii**), and in the ratio plot, values were at or above 1 throughout the range of control sizes (**Figure 8Fiv**).

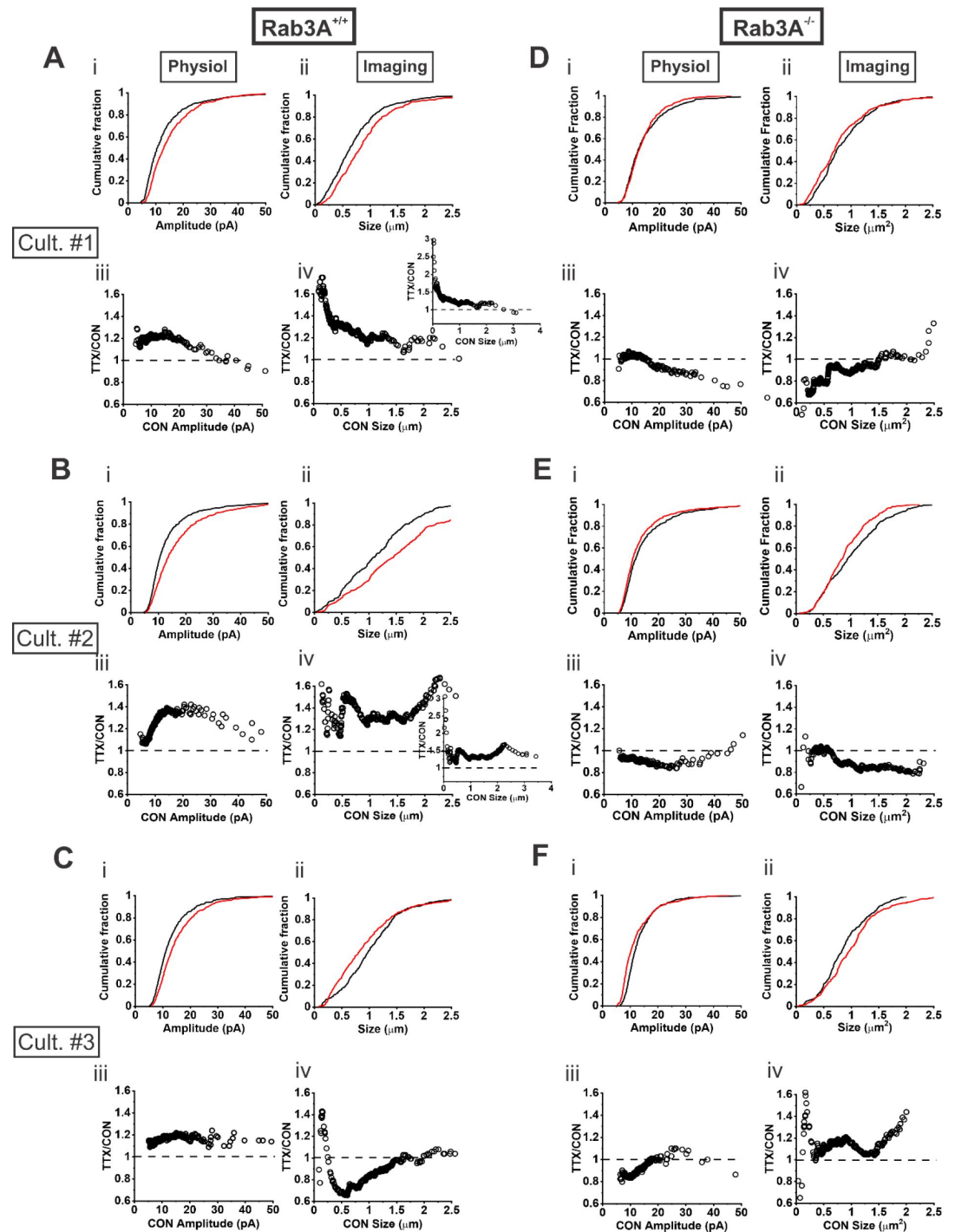


FIGURE 8

Figure 8.

mEPSC amplitudes and GluA2 receptor cluster sizes did not always change in parallel after treatment with TTX in cultures from *Rab3A*^{+/+} and *Rab3A*^{-/-} mice. A) ***Rab3A*^{+/+} Culture #1**: mEPSC amplitudes and GluA2 receptor cluster sizes increased in parallel after treatment in TTX. i) CDFs of mEPSC amplitudes of cells from untreated and TTX treated dishes were significantly different (KS test, $D = 0.16$, $p = 0.010$; CON, $n = 180$, TTX, $n = 210$). ii) CDFs of GluA2 receptor cluster sizes of dendrites from untreated and TTX treated dishes were significantly different (KS test, $D = 0.17$, $p = 3.79 \times 10^{-4}$; CON, $n = 300$, TTX, $n = 300$). iii) 35 mEPSC amplitude quantiles from each of 6 untreated cells and 30 quantiles from each of 7 TTX-treated cells were sorted from smallest to largest and used to calculate the ratio of TTX mEPSC amplitude/CON mEPSC amplitude, which was plotted vs. CON mEPSC amplitude. iv) 30 GluA2 receptor cluster size quantiles from each of 10 untreated dendrites and 30 GluA2 receptor cluster size quantiles from each of 10 TTX-treated dendrites were used

to calculate the ratio of TTX GluA2 cluster size/CON GluA2 cluster size, and plotted vs. CON GluA2 cluster size. *Inset*, full range of data. B) **Rab3A^{+/+} Culture #2**: mEPSC amplitudes and GluA2 receptor cluster sizes increased in parallel after treatment in TTX. i) CDFs of mEPSC amplitudes of cells from untreated and TTX treated dishes were significantly different (KS test, $D = 0.21$, $p = 1.06 \times 10^{-4}$; CON, $n = 210$, TTX, $n = 240$). ii) CDFs of GluA2 receptor cluster sizes of dendrites from untreated and TTX treated dishes were significantly different (KS test, $D = 0.21$, $p = 5.39 \times 10^{-6}$; CON, $n = 270$, TTX, $n = 300$). iii) 32 mEPSC amplitude quantiles from each of 7 untreated cells, and 28 mEPSC amplitude quantiles from each of 8 TTX treated cells were sorted from smallest to largest and were used to calculate the ratio of TTX mEPSC amplitude/CON mEPSC amplitude, and plotted vs. CON mEPSC amplitude. iv) 30 GluA2 receptor cluster size quantiles from each of 9 untreated dendrites and 27 GluA2 receptor cluster size quantiles from each of 10 TTX-treated dendrites were sorted from smallest to largest and used to calculate the ratio of TTX GluA2 cluster/CON GluA2 cluster size, and plotted vs. CON GluA2 cluster size. *Inset*, full range of data. C) **Rab3A^{+/+} Culture #3**: mEPSC amplitudes increased, but GluA2 receptor cluster sizes decreased, after treatment in TTX. i) CDFs of mEPSC amplitudes of cells from untreated and TTX treated dishes were significantly different (KS test, $D = 0.15$, $p = 0.002$; CON, $n = 300$, TTX, $n = 270$). ii) CDFs of GluA2 receptor cluster sizes of dendrites from untreated and TTX treated dishes were significantly different, but the TTX CDF is shifted to the *left* (KS test, $D = 0.16$, $p = 0.001$; CON, $n = 300$, TTX, $n = 270$) iii) 27 mEPSC amplitude quantiles from each of 10 untreated cells, and 30 mEPSC amplitude quantiles from each of 9 TTX-treated cells, were sorted from smallest to largest, used to calculate the ratio of TTX mEPSC amplitude/ CON mEPSC amplitude, and plotted vs. CON mEPSC amplitude. **iv)** 27 GluA2 receptor cluster size quantiles from each of 10 untreated dendrites and 30 GluA2 receptor cluster sizes from each of 9 TTX-treated dendrites were sorted from smallest to largest, used to calculate the ratio of TTX GluA2 receptor cluster size/CON GluA2 receptor cluster size, and plotted vs. CON GluA2 receptor cluster size. D) **Rab3A^{-/-} Culture #1** (note, this culture is distinct from that of Rab3A^{+/+} Culture #1). Neither mEPSC amplitudes, nor GluA2 receptor cluster size, increased following TTX treatment. i) CDFs of mEPSC amplitudes of cells from untreated and TTX-treated cultures were not significantly different (KS test, $D = 0.05$, $p = 0.90$; CON, $n = 210$, TTX, $n = 180$). ii) CDFs of GluA2 receptor cluster sizes of dendrites from untreated and TTX ([Figure 8](#) continued) treated dishes were not significantly different, although the TTX CDF is shifted slightly to the *left* (KS test, $D = 0.10$, $p = 0.08$; CON, $n = 300$, TTX, $n = 300$). iii) 30 mEPSC amplitude quantiles from each of 7 untreated cells and 35 mEPSC quantiles from each of 6 TTX-treated cells were sorted from smallest to largest, used to calculate the ratio TTX mEPSC amplitude/CON mEPSC amplitude, and plotted vs. CON mEPSC amplitude. iv) 30 GluA2 receptor cluster size quantiles from each of 10 untreated dendrites and 10 TTX-treated dendrites were sorted from smallest to largest, used to calculate the ratio TTX cluster size/CON cluster size, and plotted vs. CON cluster size. E) **Rab3A^{-/-} Culture #2** (distinct from Rab3A^{+/+} Culture #2) mEPSC amplitudes and GluA2 receptor cluster sizes decreased after TTX treatment. i) TTX mEPSC amplitudes CDF is shifted slightly to the *left* compared to the CON mEPSC amplitudes CDF but the difference is not significant (KS test, $D = 0.08$, $p = 0.28$, CON, $n = 240$, TTX, $n = 240$). ii) TTX GluA2 receptor cluster sizes CDF is significantly shifted to the *left* compared to CON GluA2 receptor cluster sizes CDF (KS test, $D = 0.13$, $p = 0.01$; CON, $n = 300$, TTX, $n = 300$). iii) 24 mEPSC amplitude quantiles from each of 8 untreated cells and 8 TTX- treated cells were sorted from smallest to largest, used to calculate the ratio TTX mEPSC amplitude/CON mEPSC amplitude, and plotted vs. CON mEPSC amplitude. iv) 27 GluA2 receptor cluster size quantiles from each of 10 untreated dendrites, and 30 GluA2 receptor cluster size quantiles from each of 9 TTX treated dendrites, were sorted from smallest to largest, used to calculate the ratio TTX GluA2 cluster size/CON cluster size, and plotted vs. CON cluster size. Both mEPSC amplitude and GluA2 receptor cluster size ratios were below 1 across the majority of data. F) **Rab3A^{-/-} Culture #3** (distinct from Rab3A^{+/+} Culture #3). mEPSC amplitudes were decreased following TTX treatment, but GluA2 receptor cluster sizes were increased. i) TTX CDF of mEPSC amplitudes was significantly shifted to the left of CON CDF (KS test, $D = 0.21$, $p = 4.85 \times 10^{-4}$, CON, $n = 180$, TTX, $n = 180$). ii) TTX CDF of GluA2 receptor cluster sizes was significantly shifted to the right of CON CDF (KS test, $D = 0.15$, $p = 0.003$, CON, $n = 300$, TTX, $n = 300$). iii) 30 mEPSC amplitude quantiles from each of 6 untreated and 6 TTX-treated cells were sorted from smallest to largest, used to calculate the ratio TTX mEPSC amplitude/CON mEPSC amplitude, and plotted vs. CON mEPSC amplitude. iv) 30 GluA2 receptor cluster size quantiles from each of 10 untreated dendrites and 10 TTX-treated dendrites were sorted from smallest to largest, used to calculate the ratio TTX GluA2 cluster size/CON GluA2 cluster size, and plotted vs. CON GluA2 cluster size.

Perhaps more striking than the broad mismatches, which could be attributed to the small sample sizes, is the complete lack of correspondence of TTX/CON ratios in the mEPSC amplitudes compared to GluA2 cluster sizes in the same cultures. In cultures from Rab3A^{+/+} mice, mEPSC amplitudes show divergent scaling similar to what we have previously reported, with the smallest CON amplitudes showing the smallest ratios, and the ratio increasing monotonically to a plateau,

or, a peak followed by a gradual decline, as CON mEPSC amplitude becomes very large. In contrast, the very smallest GluA2 cluster sizes in Rab3A^{+/+} cultures have the largest increase, with ratios as high as 3 in the two cultures that showed a broad overall increase. The ratios decline dramatically to values around 1.5, but then remain substantially above 1 throughout the entire range of data. Interestingly, the dramatic increase in the size of the smallest GluA2 clusters is absent in the two cultures from Rab3A^{-/-} mice that broadly showed no increase, suggesting this aspect is also dependent on Rab3A. It is not even possible to cut off the ratios at a certain point, to address a threshold effect such as GluA2 cluster sizes below a certain value not being measurable electrophysiologically, and get the ratios for the remaining data to match up. Taken together, these data indicate that variation in homeostatic effects on GluA2 cluster size is not responsible for the variation in homeostatic effects on mEPSC amplitude.

In summary, our matched mEPSC and receptor cluster results indicate that: 1. activity blockade resulted in an increase in mEPSC amplitudes and synaptic GluA2 receptors; 2. loss of Rab3A disrupted both the increase in mEPSC amplitudes and the increase in GluA2 receptor levels; and 3. The homeostatic effects on mEPSC amplitudes and the GluA2 cluster sizes within the same cultures differed at both the broad level, and in the specific way the ratios were non-uniform, making it difficult to conclude modulation of GluA2 cluster size is the sole determinant of homeostatic increases in mEPSC amplitude. Although it remains a possibility that the disparities in homeostatic effects on mEPSC amplitudes and GluA2 cluster size TTX/CON ratios might arise due to a sampling error, given the small number of cells sampled in each individual experiment (≤ 10), it seems unlikely that differences arising from sampling error alone would be so consistent. However, only when we are able to more easily sample larger numbers of cells, or it becomes possible to measure mEPSCs and receptor levels in the same cells, will we be able to resolve this question.

We have demonstrated here for the first time that the synaptic vesicle protein Rab3A influences the homeostatic regulation of postsynaptic GluA2 receptors. What are possible ways that Rab3A could exert a postsynaptic action? It has previously been shown that exogenous addition of TNF α to hippocampal cultures causes an increase in surface expression of GluA1 receptors (although not GluA2 receptors), and that the homeostatic increase in mEPSC amplitudes is abolished in cultures prepared from the TNF α deletion mouse (Stellwagen et al., 2005 [DOI](#); Stellwagen and Malenka, 2006 [DOI](#)). Furthermore, neurons from TNF α ^{+/+} mice plated on astrocytic feeder layers derived from TNF α ^{-/-} mice fail to show the increase in mEPSC amplitude after TTX treatment, indicating that the TNF α inducing the receptor increases following activity blockade comes from the astrocytes (Stellwagen and Malenka, 2006 [DOI](#)). Rab3A has been detected in astrocytes (Maienschein et al., 1999 [DOI](#); Hong et al., 2016 [DOI](#)), so to determine whether Rab3A is acting via regulating TNF α release from astrocytes, we performed experiments similar to those of Stellwagen and Malenka (2006 [DOI](#)) (schema illustrated in **Figure 9** [DOI](#), left side). We compared the effect of activity blockade on mEPSC amplitudes recorded from cortical neurons from Rab3A^{+/+} mice plated on Rab3A^{+/+} astrocytic feeder layers (**Figure 9A** [DOI](#)); neurons from Rab3A^{+/+} mice plated on Rab3A^{-/-} astrocytes (**Figure 9B** [DOI](#)), and neurons from Rab3A^{-/-} mice plated on Rab3A^{+/+} astrocytes (**Figure 9C** [DOI](#)). If Rab3A is required for TNF α release from astrocytes, then Rab3A^{+/+} neurons plated on Rab3A^{-/-} astrocytes should not show a homeostatic increase in mEPSC amplitude after treatment with TTX, and any cultures with Rab3A^{+/+} astrocytes should have a normal homeostatic response. We found the opposite result: mEPSC amplitudes increased dramatically in cultures where Rab3A was present in neurons (**Figures 9A and B** [DOI](#)), but much more modestly increased in cultures where Rab3A was present only in astrocytes (**Figure 9C** [DOI](#)).

Interestingly, the homeostatic effects appear to be larger in the feeder layer cultures, compared to the effects observed in our previous neuronal/astrocyte mixed cultures prepared directly onto poly-L-lysine coated coverslips, but the ratio plots clearly show scaling was divergent under both plating conditions (compare **Figures 1Giii** [DOI](#) with **Figures 9Aiii** [DOI](#) and **Biii**). We noticed that in the astrocyte feeder layer cultures, and in the matched mEPSC amplitude and GluA2 receptor

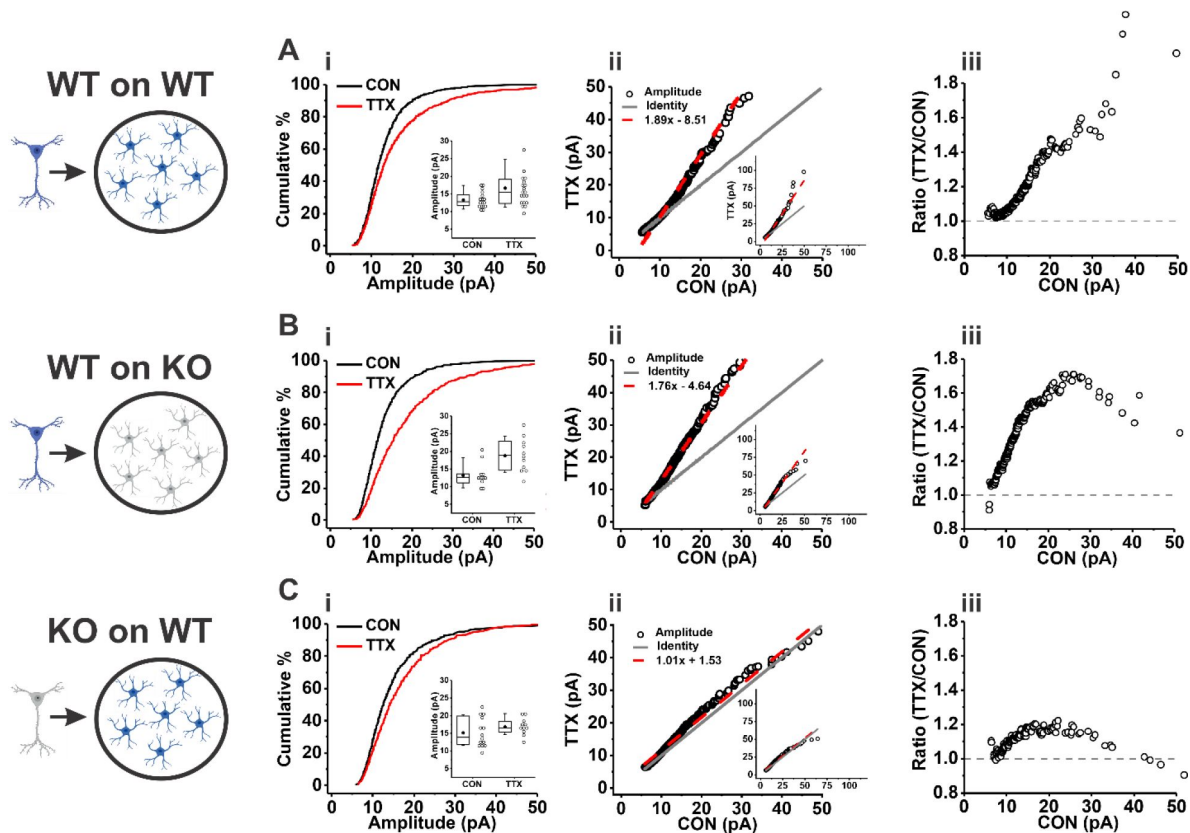


Figure 9.

Rab3A in neurons, not astrocytes, was required for full homeostatic plasticity after activity blockade. A-C, Data from dissociated cortical neurons plated on an astrocyte feeder layer, each prepared separately from the type of mice depicted in the schema at left: A) Neurons from Rab3A^{+/+} mice plated on astrocytes from Rab3A^{+/+} mice; i) CDFs for mEPSC amplitudes from untreated (CON, black curve) and TTX-treated (TTX, red curve) cultures (KS test, test statistic, $D = 0.141$, $p = 2.74 \times 10^{-5}$, $n = 510$ CON, 600 TTX samples; *inset*, means, CON, 13.3 ± 0.5 pA, $n = 17$ cells, TTX, 16.7 ± 1.2 pA, $n = 20$ cells, $p = 0.03$, Kruskal Wallis test; ii) rank order plot and linear regression fit, slope = 1.89, intercept = -8.51, $R^2 = 0.97$; iii) ratio plot of TTX mEPSC amplitude/CON amplitude vs. CON amplitude; B) Neurons from Rab3A^{+/+} mice plated on astrocytes from Rab3A^{-/-} mice; i) CDFs of mEPSC amplitudes, CON, black curve; TTX, red curve (KS test, test statistic 0.273, $p = 2.98 \times 10^{-11}$, $n = 330$ CON, 330 TTX samples; *inset*, means, CON, 13.3 ± 1.0 pA, $n = 11$ cells, TTX, 18.8 ± 1.4 pA, $n = 11$ cells, $p = 0.005$, Kruskal-Wallis test; ii) rank order plot and linear regression fit, slope = 1.76, intercept = -4.64, $R^2 = 0.98$; iii) ratio plot of TTX mEPSC amplitude/TTX mEPSC amplitude vs. CON mEPSC amplitude; C) Neurons from Rab3A^{-/-} neurons plated on astrocytes from Rab3A^{+/+} mice; i) CDFs of mEPSC amplitudes, CON, black curve, TTX, red curve (KS test, test statistic 0.126, $p = 0.005$, $n = 420$ CON, 330 TTX samples; *inset*, means, CON, 15.2 ± 1.1 pA, $n = 14$ cells, TTX, 16.9 ± 0.7 pA, $n = 11$ cells, $p = 0.125$, Kruskal-Wallis test; ii) rank order plot and linear regression fit; slope = 1.01, intercept = 1.53, $R^2 = 0.97$; ratio plot, TTX mEPSC amplitude/CON mEPSC amplitude vs. CON mEPSC amplitude. Gray line, identity; red dashed line, linear regression fit. Insets in rank order plots depict full range of data.

measurement experiments, the mean mEPSC amplitude in the untreated cultures prepared from Rab3A^{-/-} mice was slightly, but not significantly, larger, compared to the mean mEPSC amplitude for untreated cultures prepared from Rab3A^{+/+} mice in the same experiments (astrocyte feeder layer, Rab3A^{+/+} neurons on Rab2A^{+/+} astrocytes, 13.3 ± 0.5 pA, Rab3A^{-/-} on Rab3A^{+/+} astrocytes, 15.2 ± 1.1 pA; matched mEPSC and receptor experiments, Rab3A^{+/+} 13.7 ± 4.5 pA, Rab3A^{-/-} 14.9 ± 3.8 pA. It is therefore possible that loss of Rab3A, like expression of the Rab3A Earlybird mutant, affects basal mEPSC amplitude, albeit to a lesser extent. However, it could also be that these differences reflect random variation in mEPSC amplitude from culture to culture.

In summary, in the absence of Rab3A from astrocytes, the divergent scaling of mEPSC amplitude following activity blockade was completely normal, whereas in the absence of Rab3A in neurons, scaling was greatly diminished. This result makes it highly unlikely that Rab3A is required for the release of TNFα, or another factor from astrocytes, that induces a homeostatic upregulation of postsynaptic receptors and thereby increases mEPSC amplitude following TTX treatment. Neuronal Rab3A appears to mediate the homeostatic increase in mEPSC amplitude following activity blockade.

Discussion

We found that homeostatic synaptic plasticity of mEPSC amplitude in dissociated mixed cultures of mouse cortical neurons and astrocytes behaved remarkably similar to the mouse NMJ in response to loss of Rab3A function: scaling up of mEPSC amplitude following prolonged network silencing by TTX was strongly diminished in cultures from Rab3A^{-/-}, and in cultures from Rab3A^{Ebd/Ebd} mice, basal mEPSC amplitude was increased compared to that of wild-type cultures and was not further modified following 48-hour treatment with TTX. These results suggest that normal function of the presynaptic vesicle protein Rab3A is required for the homeostatic scaling up of mEPSC amplitude in cortical cultures and at the NMJ in vivo.

Rab3A not likely to regulate receptor trafficking

We demonstrated that an increase in synaptic GluA2 receptors accompanies the increase in mEPSC amplitude in cultures of dissociated mouse cortical neurons after 48-hour TTX treatment. By examining the same cultures with both mEPSC and immunofluorescence measurements, we found: 1. the effect on GluA2 receptor levels, whether measured as the size of a synaptic cluster or the density of receptors in a cluster, was less robust than the effect on mEPSC amplitudes; 2. the homeostatic effect on mEPSC amplitudes and GluA2 receptors substantially differed from each other both broadly (mEPSC amplitudes could increase while GluA2 cluster sizes decreased, and vice versa), and in the details of non-uniform scaling, one example being that the smallest mEPSC amplitudes showed the least effects but the smallest GluA2 clusters showed the greatest effects. Interestingly, there are other examples where the effect of activity blockade by either TTX alone (Hu et al., 2010 [\[1\]](#)), TTX and AP5 (Letellier et al., 2014 [\[2\]](#)), or DNQX (Blackman et al., 2012 [\[3\]](#)) on AMPAR levels did not match that of mEPSC amplitudes, although in these cases, the increases in receptors were larger than the increases of mEPSC amplitudes. The striking differences between TTX effects on AMPARs and mEPSC amplitudes indicates that there may be another factor contributing to mEPSC amplitude besides levels of AMPAR expression. One possibility is alterations in the type of AMPA receptor expressed, since GluA1 homomers have much greater conductance than GluA2-containing receptors (Oh and Derkach, 2005 [\[4\]](#); Benke and Traynelis, 2019 [\[5\]](#)). Increases in mEPSC amplitudes not accompanied by increases in receptor numbers have been attributed to such a switch (Hou et al., 2015 [\[6\]](#); Silva et al., 2019 [\[7\]](#); Dubes et al., 2022 [\[8\]](#)).

In addition to finding a disparity between synaptic GluA2 receptor and mEPSC responses to prolonged activity blockade, there are two other reasons we do not think Rab3A impacts homeostatic synaptic plasticity solely through postsynaptic regulation of GluA2 receptor

trafficking to the plasma membrane. First, prolonged activity blockade with TTX at the NMJ led to an increase in mEPSC amplitude that was not accompanied by an increase in AChR levels, yet it was dependent on Rab3A (Wang et al., 2011 [DOI](#)). Second, for Rab3A to modulate AMPAR trafficking, it must be located in the postsynaptic dendrite. Multiple presynaptic molecules, such as SNARE proteins, synaptotagmins, and NSF have been identified in the postsynaptic compartment (Lledo et al., 1998 [DOI](#); Nishimune et al., 1998 [DOI](#); Osten et al., 1998 [DOI](#); Song et al., 1998 [DOI](#); Araki et al., 2010 [DOI](#); Kennedy et al., 2010 [DOI](#); Suh et al., 2010 [DOI](#); Jurado et al., 2013 [DOI](#); Hussain and Davanger, 2015 [DOI](#); Gu et al., 2016 [DOI](#); Wu et al., 2017 [DOI](#)), but to our knowledge, Rab3A has not.

Rab3A may regulate quantal size presynaptically

Based on our findings, the lack of postsynaptic Rab3A expression and the known presynaptic expression of Rab3A, Rab3A may regulate mEPSC amplitude presynaptically by modulating: a) the amount of transmitter transported into the vesicle, b) the size of the vesicle, or c) fusion pore characteristics. mRNA for the glutamate transporter, VGLUT1, was increased in rat cortical cultures after 48-hr treatment with TTX (De Gois et al., 2005 [DOI](#)), and VGLUT1 relative to synapsin I in rat hippocampal cultures was increased after 48-hr treatment with AP-5 (Wilson et al., 2005 [DOI](#)). This activity-dependent regulation of VGLUT1 was recently corroborated in a proteome study (Dorrbau et al., 2020 [DOI](#)), and it has also been shown that mEPSC amplitudes are increased after overexpression of VGLUT at the *Drosophila* NMJ (Daniels et al., 2004 [DOI](#)) and in rat hippocampal cultures (Wilson et al., 2005 [DOI](#)). These data suggest that increases in VGLUT1 after activity-blockade could contribute to the homeostatic increase in mEPSC amplitude. Although there is no direct evidence for an interaction between VGLUT1 and Rab3A, Rab3A levels are reduced 50% in hippocampal extracts from mice lacking VGLUT1 (Fremeau et al., 2004). A Rab3A-dependent modulation of presynaptic quantal size would explain the homeostatic effects on mEPSC amplitudes that are not matched by changes in receptor levels.

Interestingly, Rab3A plays a role in regulating vesicle size. In adrenal chromaffin cells from mice lacking all 4 Rab3 isoforms (A, B, C and D) or mice heterozygous for Rab3A and lacking the B, C, and D isoforms (Rab3A^{+/-}BCD^{-/-}) with only 1 copy of Rab3A, there was a 20% increase in large dense core vesicle (DCV) diameter that did not reach significance ($p = 0.11$) (Schonn et al., 2010 [DOI](#)). Similarly, in the exocrine pancreas and parotid gland of the Rab3D^{-/-}, large DCV diameter was increased by 22% (Riedel et al., 2002 [DOI](#)). This magnitude of change in diameter would produce a greater than 80% increase in volume. Also, an increase in volume has been shown to increase miniature endplate junctional current amplitude at the *Drosophila* NMJ (Karunanithi et al., 2002 [DOI](#)). There have not been any reports that small synaptic vesicle size is increased in Rab3^{-/-} mice, but small synaptic vesicle size is regulated by activity: vesicle size increased in retinotectal terminals within the optic tectum following enucleation in pigeon (Cuenod et al., 1970 [DOI](#)), with similar results in cat and monkey (Lloret and Saavedra, 1975 [DOI](#)), in rat caudate nucleus following cortical ablation (Kawana et al., 1971 [DOI](#)), and in rat ventral spinal cord following immobilization (Cheresharov et al., 1978 [DOI](#)).

Finally, we previously showed that at the mouse NMJ, both loss of Rab3A and activity blockade by TTX caused an increase in the frequency of abnormally slow rising and/or prolonged duration mEPSCs (Wang et al., 2011 [DOI](#)). We further demonstrated that the loss of Rab3A increased the frequency of very small amplitude fusion pore feet in mouse chromaffin cells (Wang et al., 2008 [DOI](#)). Taken together, these data suggested that Rab3A acts to prevent abnormal fusion pore release events. However, at the NMJ and in chromaffin cells, the kinetics (rise time, decay, half width) of the vast majority of release events were unchanged in the absence of Rab3A, making it unlikely that the shift in the distribution of mEPSC amplitudes is mediated by a Rab3A-dependent change in fusion pore kinetics.

Neuronal, not astrocytic, Rab3A may be required for the homeostatic release of a signaling molecule

Our results support the surprising possibility that presynaptic Rab3A regulates postsynaptic AMPARs, rather than occurring through a direct effect on the trafficking of AMPARs by Rab3A in the dendrite. Astrocytic release of TNF α was shown to mediate the increase in AMPARs in prolonged activity-blocked hippocampal cultures (Stellwagen and Malenka, 2006 [DOI](#)), but we found that loss of Rab3A in astrocytes does not disrupt the increase in mEPSC after activity blockade. Further, we previously showed that the homeostatic increase in NMJ mEPC amplitude was completely normal in the absence of TNF α (Wang et al., 2011 [DOI](#)).

Another possible way Rab3A might mediate regulation of postsynaptic AMPARs is through the presynaptic release of a signaling molecule, such as brain-derived neurotrophic factor (BDNF), that acts anterogradely to alter postsynaptic AMPAR levels. Addition of exogenous BDNF to neuronal cultures prevents the increase in mEPSC amplitude following activity blockade (Rutherford et al., 1998 [DOI](#); Benevento et al., 2016 [DOI](#)) but see (Smith-Dijak et al., 2019 [DOI](#)), and BDNF mRNA is reduced after activity blockade in vitro (Benevento et al., 2016 [DOI](#); Miyasaka and Yamamoto, 2021 [DOI](#)) and in vivo (Castren et al., 1992 [DOI](#)). Also, after reduction of secreted BDNF in culture media via the BDNF scavengers TrkB-FC or TrkB-IgG, mEPSC amplitudes are increased (Rutherford et al., 1998 [DOI](#); Benevento et al., 2016 [DOI](#); Smith-Dijak et al., 2019 [DOI](#)). Rab3B, a Rab3 family member with strong homology to Rab3A but expressed more highly in inhibitory nerve terminals (Tsetsenis et al., 2011 [DOI](#)), is modified epigenetically following TTX treatment by the same DNA methylation pathway as that for BDNF (Benevento et al., 2016 [DOI](#)). In addition, reduction of Rab3A in astrocytes decreases astrocytic BDNF release (Hong et al., 2016 [DOI](#)), and treatment of cultures with BDNF can increase Rab3A levels (Takei et al., 1997 [DOI](#); Thakker-Varia et al., 2001 [DOI](#)) but see (Shinoda et al., 2014 [DOI](#)). Taken together, these results suggest an interplay between BDNF and Rab3A in the activity-dependent regulation of AMPA receptors.

Rab3A is required for divergent scaling

We have established that homeostatic synaptic plasticity in mouse cortical cultures, rat cortical cultures, and mouse hippocampal cultures demonstrates divergent, rather than uniform, scaling (Hanes et al., 2020 [DOI](#); Koesters et al., 2022 [DOI](#)). The deviation from uniformity suggests that individual synapses may be independently regulated, and do not require a global, cell-wide signal. Our conclusion is supported by multiple studies demonstrating local activity-dependent regulation of mEPSC amplitudes at subsets of synapses (Ju et al., 2004 [DOI](#); Sutton et al., 2006 [DOI](#); Hou et al., 2008 [DOI](#); Beique et al., 2011 [DOI](#); Hou et al., 2011 [DOI](#); Letellier et al., 2014 [DOI](#)). It is further supported by the recent finding that in hippocampal cultures treated with TTX, AMPA receptors increase selectively at synaptopodin-positive sites, which are already larger and contain higher amounts of AMPA receptors (Dubes et al., 2022 [DOI](#)). The authors provide evidence that the process depends on the microRNA miR-124: miR-124 is reduced after activity blockade; and overexpression of miR-124 blocks the TTX-induced increase in AMPA receptors and mEPSC amplitudes. Increases in miR-124 have also been linked to decreases in mEPC amplitudes at the NMJs of the slow channel syndrome mouse (mSCS) in which prolongation of ACh currents causes excess excitation (Zhu et al., 2013 [DOI](#)). Zhu and colleagues provide evidence for a postsynaptic calpain-Cdk5-nNOS pathway that is activated by increased calcium levels, with the NO feeding back retrogradely to regulate miR-124, which in turn directly interacts with Rab3A to affect quantal content (Zhu et al., 2013 [DOI](#)), but it is also possible that Rab3A is contributing to the mEPSC amplitude changes in mSCS. In sum, these studies suggest miR-124 could tie together presynaptic changes (Rab3A, transmitter released from a single vesicle) and postsynaptic changes (synaptopodin, AMPA receptors) in divergent homeostatic synaptic plasticity.

A model for the role of Rab3A in homeostatic plasticity of mEPSC amplitude

We propose a model for Rab3A function in homeostatic synaptic plasticity of mEPSC amplitude in which Rab3A contributes to presynaptic quantal size through the amount of transmitter released by fusion of a single vesicle, and to postsynaptic quantal size through the number of postsynaptic receptors. Normally, Rab3A cycles between GTP and GDP-bound forms (“State 1,” indicated by red tag in [Figure 10A](#)) and promotes synaptic vesicle mobilization and fusion, at the same time maintaining vesicle size/transmitter content within a narrow range. In this normally cycling Rab3A state, an anterograde signal dependent on Rab3A maintains GluA2 receptors at their normal, restricted level. Activity blockade leads to a buildup of an alternate form of Rab3A, possibly one that is unable to cycle through its GTP and GDP-bound forms (“State 2,” indicated by gray tag in [Figure 10B](#)). Without the quality control mechanism conferred by State 1 Rab3A, vesicle size/transmitter content skews to higher values, and the anterograde signal is abnormal, leading to GluA2 receptors levels rising outside normal limits. Not shown here, the Earlybird point mutation may permanently be in “State 2,” as the mutation is in the guanine nucleotide binding region (Kapfhamer), which would lead to increased transmitter released from single vesicles and increased receptor levels.

We have demonstrated an essential role of the presynaptic vesicle protein Rab3A in the homeostatic increase in mEPSC amplitude and GluA2 AMPA receptor levels at synaptic sites in mouse dissociated cortical cultures. To our knowledge, this is the first evidence of a presynaptic protein being implicated in the homeostatic regulation of mEPSC amplitude and AMPARs in neuronal cultures. The current results extend our previous findings that showed the homeostatic increase in mEPSC amplitudes at the mouse NMJ *in vivo* depends on Rab3A. Under conditions where postsynaptic receptors are close to being saturated by the amount of transmitter in a single vesicle (possibly in cortical cultures but not at the NMJ), it may be necessary to increase both the amount of transmitter released and the number or function of postsynaptic receptors in order to observe a homeostatic increase in the physiological signal, mEPSC amplitude. Our results suggest that in cortical cultures, neuronal Rab3A is a key player regulating the homeostatic increase in synaptic strength on both sides of the synapse.

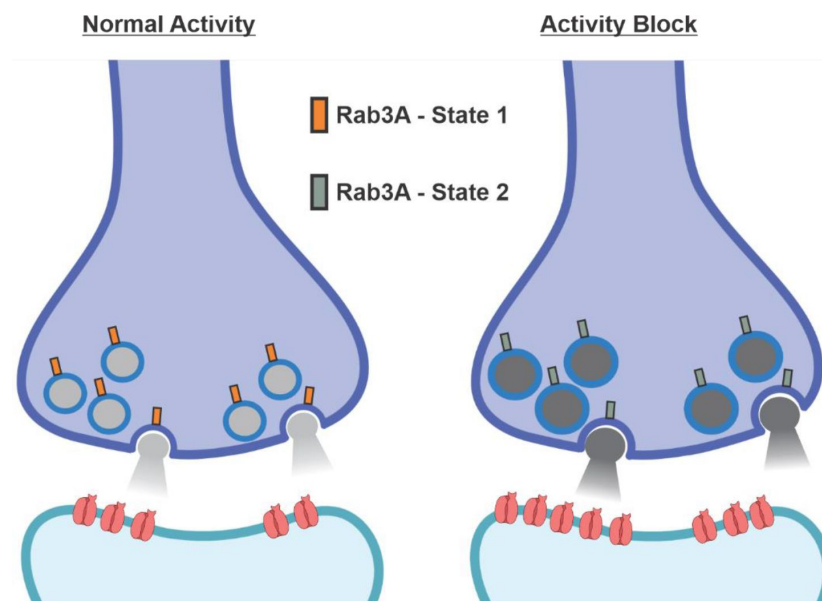


Figure 10.

Rab3A is required for increased mEPSC amplitude after activity blockade via 2 mechanisms: 1. increased amount of transmitter released from a single vesicle and 2. increased postsynaptic receptors. *Left:* in the presence of normal activity, Rab3A is in its control state ("State 1," red), maintaining mEPSC amplitude within a limited range by regulating either vesicle size or transmitter content, or both, and by regulating the number of postsynaptic receptors, likely through release of an anterograde signal. *Right:* after activity is blocked, Rab3A accumulates in an alternate functional state ("State 2," gray) resulting in quantal size expanding to larger ranges because of increased vesicle size or transmitter content and because of increased numbers of receptors no longer being under the normal control of the anterograde signal.

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Reviewer #1 (Public Review):

Koesters and colleagues investigated the role of the presynaptic small GTPase Rab3A in homeostatic scaling of miniature synaptic transmission in primary mouse cortical cultures using electrophysiology and immunohistochemistry. The major finding is that TTX incubation for 48 hours does not induce an increase in the amplitude of excitatory synaptic miniature events in neuronal cultures derived from Rab3A KO and Rab3A Earlybird mutant mice. NASPM application had comparable effects on mEPSC amplitude in control and after TTX, implying that Ca²⁺-permeable glutamate receptors are unlikely modulated during synaptic scaling. Immunohistochemical analysis revealed an increase in GluA2 puncta size and intensity in wild type, but not Rab3A KO cultures. Finally, they provide evidence that loss of Rab3A in neurons, but not astrocytes, blocks homeostatic scaling. Based on these data, the authors propose a model in which presynaptic Rab3A is required for homeostatic scaling of synaptic transmission through GluA2-dependent and independent mechanisms.

While the title of the manuscript is mostly supported by data of solid quality, many conclusions, as well as the final model, cannot be derived from the results presented. Importantly, the results do not indicate that Rab3A modulates quantal size on both sides of the synapse. Moreover, several analysis approaches seem inappropriate.

The following points should be addressed:

1. The model shown in Figure 10 is not supported by the data. The authors neither provide evidence for two different functional states of Rab3A being involved in mEPSC amplitude modulation, nor for a change in glutamate content of vesicles. Furthermore, the data do not fully support the conclusion of a presynaptic role for Rab3A in homeostatic scaling.
2. The analysis of mEPSC data using quantile sampling followed by ratio calculation is not meaningful under the tested experimental conditions because of the following reasons: (i) The analysis implicitly assumes that all events have been detected. The prominent mEPSC frequency increase after TTX suggests that this is not the case, i.e., many (small) mEPSCs are likely missed under control conditions. (ii) The analysis is used to conclude how events of a certain size are altered by TTX treatment. However, this analysis compares the smallest mEPSCs of the TTX condition with the smallest control mEPSCs, but this is not a pre-post experimental design. Variation between cells and between coverslips will markedly affect the results and lead to misleading interpretations. (iii) The ratio (TTX/control) vs. control plots seem to suffer from a division by small value artifact (see Figure 6F). Correspondingly, ratio-analysis differs considerably for different control conditions (Fig. 1Giii, Fig. 2Giii, Fig. 6C, Fig. 9A).
3. As noted by the authors in a previous publication (Hanes et al. 2020), statistical analysis of CDFs suffers from n-inflation. In addition, the quantile sampling method chosen violates an important assumption of the K-S test. Indeed, p-values for these comparisons are typically several orders of magnitude smaller. Given that the statistical N most likely corresponds to the number of cultures (see, e.g., <https://doi.org/10.1371/journal.pbio.2005282>), CDF comparisons are not informative and should thus not be used to draw conclusions from the data. The plots can be informative, though.
4. How does recoding noise and the mEPSC amplitude threshold affect "divergent scaling"?
5. What is the justification for the line fits of the ratio data/how was the fit range chosen?
6. TTX application induces a significant increase in mEPSC amplitude in Rab3A^{-/-} mice in two out of three data sets (Figs. 1 and 9). Hence, the major conclusion that Rab3A is required for homeostatic scaling is only partially supported by the data.

7. Line 289: A comparison of p-values between conditions does not allow any meaningful conclusions.
8. There is a significant increase in baseline mEPSC amplitude in Rab3AEbd/Ebd (15 pA) vs. Rab3AEbd/+ (11 pA) cultures, but not in Rab3A-/- (13.6 pA) vs. Rab3A+/+ (13.9 pA). Although the nature of scaling was different between Rab3AEbd/Ebd vs. Rab3AEbd/+, and Rab3AEbd/Ebd with vs. without TTX, the question arises whether the increase in mEPSC amplitude in Rab3AEbd/Ebd is Rab3A dependent. Could a Rab3A independent mechanism occlude scaling?
9. Figure 4: NASPM appears to have a stronger effect on mEPSC frequency in the TTX condition vs. control (-40% vs. -15%). A larger sample size might be necessary to draw definitive conclusions on the contribution of Ca²⁺-permeable AMPARs.
10. The authors discuss previous papers showing changes in VGLUT1 intensity. Was VGLUT1 intensity altered in the stainings presented in the manuscript?
11. The change in GluA2 area or fluorescence intensity upon TTX treatment in controls is modest. How does the GluA2 integral change?
12. The quantitative comparison between physiology and microscopy data is problematic. The authors report a mismatch in ratio values between the smallest mEPSC amplitudes and smallest GluA2 receptor cluster sizes (l. 464; Figure 8). Is this comparison affected by the fluorescence intensity threshold? What was the rationale for a threshold of 400 a.u. or 450 a.u.? How does this threshold compare to the mEPSC threshold of 3 pA? The conclusion that an increase in AMPAR levels is not fully responsible for the observed mEPSC increase is mainly based on the rank-order analysis of GluA2 intensity, yielding a slope of ~0.9. There are several points to consider here: (i) GluA2 fluorescence intensity did increase on average, as did GluA2 cluster size. (ii) The increase in GluA2 cluster size is very similar to the increase in mEPSC amplitude (each approx. 18-20%). (iii) Are there any reports that fluorescence intensity values are linearly reporting mEPSC amplitudes (in this system)? Antibody labelling efficiency, and false negatives of mEPSC recordings may influence the results. The latter was already noted by the authors. (iv) It is not entirely clear if their imaging experiments will sample from all synapses. Other AMPAR subtypes than GluA2 could contribute, as could kainate or NMDA receptors. Furthermore, the statement "complete lack of correspondence of TTX/CON ratios" is not supported by the data presented (l. 515ff). First, under the assumption that no scaling occurs in Rab3A-/-, the TTX/CON ratios show a 20-30% change, which indicates the variation of this readout. Second, the two examples shown in Figure 8 for Rab3A+/+ are actually quite similar (culture #1 and #2), particularly when ignoring the leftmost section of the data, which is heavily affected by the raw values approaching zero.
13. Figure 7A: TTX CDF was shifted to smaller mEPSC amplitude values in Rab3A-/- cultures. How can this be explained?

Reviewer #2 (Public Review):

In this study, Koesters et al. investigated whether Rab3A, a small GTPase that regulates synaptic vesicle fusion pore opening, is required for excitatory synaptic scaling in response to TTX-induced activity suppression in dissociated mouse cortical neuronal culture. They first show that, while pyramidal neurons from wild-type (WT) littermates show normal synaptic scaling in response to 48h of TTX treatment (~30% increase in the mean mEPSC amplitude), those from two different mouse lines with either deletion (Rab3A-/-) or loss-of-function mutation of Rab3A (Rab3AEbd/Ebd) fail to engage this homeostatic compensation. They perform cumulative distribution analysis to show that the mEPSC population has gone through divergent scaling in WT neurons. Similarly, this phenomenon is absent in neurons from the two Rab3A mouse lines. They further demonstrate that GluA2-containing AMPARs likely account for the increase in mEPSC amplitudes by comparing measurements before and after washing in blockers specific for GluA2-lacking AMPARs. Subsequently, they perform electrophysiology and immunohistochemistry side by side for WT neurons from the same culture following TTX treatment, and find that both mEPSC amplitudes and GluA2 cluster

sizes have shifted towards higher values, while GluA2 cluster intensity remains unchanged. Importantly, all these homeostatic compensations are absent in Rab3A^{-/-} neurons. Finally, they mix neurons and astrocyte feeders either from WT or Rab3A^{-/-} mice, which reveals that neuronal but not astrocytic Rab3A knockout leads to impaired scaling up of mEPSCs. They conclude that Rab3A is required for homeostatic scaling up of mEPSC amplitude in cortical neurons, most likely from the presynaptic side.

Although the authors have raised an interesting question, their conclusion is not well supported by the data presented. I list my technical and conceptual concerns below.

Technical concerns:

1. The culture condition is questionable. The authors saw no NMDAR current present during spontaneous recordings, which is worrisome since NMDARs should be active in cultures with normal network activity (Watt et al., 2000; Sutton et al., 2006). It is important to ensure there is enough spiking activity before doing any activity manipulation. Similarly, it is also unknown whether spiking activity is normal in Rab3A KO/Ebd neurons.
2. Selection of mEPSC events is not conducted in an unbiased manner. Manually selecting events is insufficient for cumulative distribution analysis, where small biases could skew the entire distribution. Since the authors claim their ratio plot is a better method to detect the uniformity of scaling than the well-established rank-order plot, it is important to use an unbiased population to substantiate this claim.
3. Immunohistochemistry data analysis is problematic. The authors only labeled dendrites without doing cell-fills to look at morphology, so it is questionable how they differentiate branches from pyramidal neurons and interneurons. Since glutamatergic synapses on these two types of neuron scale in the opposite directions, it is crucial to show that only pyramidal neurons are included for analysis.

Conceptual concerns:

The only novel finding here is the implicated role for Rab3A in synaptic scaling, but insights into mechanisms behind this observation are lacking. The author claims that Rab3A likely regulates scaling from the presynaptic side, yet there is no direct evidence from data presented. In its current form, this study's contribution to the field is very limited.

1. Their major argument for this is that homeostatic effects on mEPSC amplitudes and GluA2 cluster sizes do not match. This is inconsistent with reports from multiple labs showing that upscaling of mEPSC amplitude and GluA2 accumulation occur side by side during scaling (Ibata et al., 2008; Pozo et al., 2012; Tan et al., 2015; Silva et al., 2019). Further, because the acquisition and quantification methods for mEPSC recordings and immunohistochemistry imaging are entirely different (each with its own limitations in signal detection), it is not convincing that the lack of proportional changes must signify a presynaptic component.
2. The authors also speculate in the discussion that presynaptic Rab3A could be interacting with retrograde BDNF signaling to regulate postsynaptic AMPARs. Without data showing Rab3A-dependent presynaptic changes after TTX treatment, this argument is not compelling. In this retrograde pathway, BDNF is synthesized in and released from dendrites (Jakawich et al., 2010; Thapliyal et al., 2022), and it is entirely possible for postsynaptic Rab3A to interfere with this process cell-autonomously.
3. The authors propose that a change in AMPAR subunit composition from GluA2-containing ones to GluA1 homomers may account for the distinct changes in mEPSC amplitudes and GluA2 clusters. However, their data from the Naspm wash-in experiments clearly show that GluA1 homomer contributions have not changed before and after TTX treatment.

Ibata K, Sun Q, Turrigiano GG (2008) Rapid synaptic scaling induced by changes in postsynaptic firing. *Neuron* 57:819-826.

Jakawich SK, Nasser HB, Strong MJ, McCartney AJ, Perez AS, Rakesh N, Carruthers CJL, Sutton MA (2010) Local Presynaptic Activity Gates Homeostatic Changes in Presynaptic Function Driven by Dendritic BDNF Synthesis. *Neuron* 68:1143-1158.

Pozo K, Cingolani LA, Bassani S, Laurent F, Passafaro M, Goda Y (2012) $\beta 3$ integrin interacts directly with GluA2 AMPA receptor subunit and regulates AMPA receptor expression in hippocampal neurons. *Proceedings of the National Academy of Sciences* 109:1323-1328.

Silva MM, Rodrigues B, Fernandes J, Santos SD, Carreto L, Santos MAS, Pinheiro P, Carvalho AL (2019) MicroRNA-186-5p controls GluA2 surface expression and synaptic scaling in hippocampal neurons. *Proceedings of the National Academy of Sciences* 116:5727-5736.

Sutton MA, Ito HT, Cressy P, Kempf C, Woo JC, Schuman EM (2006) Miniature Neurotransmission Stabilizes Synaptic Function via Tonic Suppression of Local Dendritic Protein Synthesis. *Cell* 125:785-799.

Tan HL, Queenan BN, Hugarir RL (2015) GRIP1 is required for homeostatic regulation of AMPAR trafficking. *Proceedings of the National Academy of Sciences* 112:10026-10031.

Thapliyal S, Arendt KL, Lau AG, Chen L (2022) Retinoic acid-gated BDNF synthesis in neuronal dendrites drives presynaptic homeostatic plasticity. *eLife* 11:e79863.

Watt AJ, Rossum MCW van, MacLeod KM, Nelson SB, Turrigiano GG (2000) Activity Coregulates Quantal AMPA and NMDA Currents at Neocortical Synapses. *Neuron* 26:659-670.

Reviewer #3 (Public Review):

Summary: The authors clearly demonstrate the Rab3A plays a role in HSP at excitatory synapses, with substantially less plasticity occurring in the Rab3A KO neurons. There is also no apparent HSP in the Earlybird Rab3A mutation, although baseline synaptic strength seems already elevated. In this context, it is unclear if the plasticity is absent or just occluded by a ceiling effect due the synapses already being strengthened. The authors do appropriately discuss both options. There are also differences in genetic background between the Rab3A KO and Earlybird mutants that could also impact the results, which are also noted. The authors have solid data showing that Rab3A is unlikely to be active in astrocytes. Finally, they attempt to study the linkage between synaptic strength during HSP and AMPA receptor trafficking, and conclude that trafficking is largely not responsible for the changes in synaptic strength.

Strengths: This work adds another player into the mechanisms underlying an important form of synaptic plasticity. The plasticity is only reduced, suggesting Rab3A is only partially required and perhaps multiple mechanisms contribute. The authors speculate about some possible novel mechanisms.

Weaknesses: However, the rather strong conclusions on the dissociation of AMPAR trafficking and synaptic response are made from somewhat weaker data. The key issue is the GluA2 immunostaining in comparison with the mEPSC recordings. Their imaging method involves only assessing puncta clearly associated with a MAP2 labeled dendrite. This is a small subset of synapses, judging from the sample micrographs (Fig 5). To my knowledge, this is a new and unvalidated approach that could represent a particular subset of synapses not representative of the synapses contributing to the mEPSC change (they are also sampling different neurons for the two measurements; an additional unknown detail is how far from the cell body were the analyzed dendrites for immunostaining). While the authors acknowledge that a sampling issue could explain the data, they still use this data to draw strong conclusions about the lack

of AMPAR trafficking contribution to the mEPSC amplitude change. This apparent difference may be a methodological issue rather than a biological one, and at this point it is impossible to differentiate these. It will unfortunately be difficult to validate their approach. Perhaps if they were to drive NMDA-dependent LTD or chemLTP, and show alignment of the imaging and ephys, that would help. More helpful would be recordings and imaging from the same neurons but this is challenging. Sampling from identified synapses would of course be ideal, perhaps from 2P uncaging combined with SEP-labeled AMPARs, but this is more challenging still. But without data to validate the method, it seems unwarranted to make such strong conclusions such as that AMPAR trafficking does not underlie the increase in mEPSC amplitude, given the previous data supporting such a model.

Other questions arise from the NASPM experiments, used to justify looking at GluA2 (and not GluA1) in the immunostaining. First, there is a frequency effect that is quite unclear in origin. One would expect NASPM to merely block some fraction of the post-synaptic current, and not affect pre-synaptic release or block whole synapses. It is also unclear why the authors argue this proves that the NASPM was at an effective concentration (lines 399-400). Further, the amplitude data show a strong trend towards smaller amplitude. The p value for both control and TTX neurons was 0.08 - it is very difficult to argue that there is no effect. And the decrease is larger in the TTX neurons. Considering the strong claims for a pre-synaptic locus and the use of this data to justify only looking at GluA2 by immunostaining, these data do not offer much support of the conclusions. Between the sampling issues and perhaps looking at the wrong GluA subunit, it seems premature to argue that trafficking is not a contributor to the mEPSC amplitude change, especially given the substantial support for that hypothesis. Further, even if trafficking is not the major contributor, there could be shifts in conductance (perhaps due to regulation of auxiliary subunits) that does not necessitate a pre-synaptic locus. While the authors are free to hypothesize such a mechanism, it would be prudent to acknowledge other options and explanations.

The frequency data are missing from the paper, with the exception of the NASPM dataset. The mEPSC frequencies should be reported for all experiments, particularly given that Rab3A is generally viewed as a pre-synaptic protein regulating release. Also, in the NASPM experiments, the average frequency is much higher in the TTX treated cultures. Is this statistically above control values?

Unaddressed issues that would greatly increase the impact of the paper:

1. Is Rab3A acting pre-synaptically, post-synaptically or both? The authors provide good evidence that Rab3A is acting within neurons and not astrocytes. But where it is acting (pre or post) would aid substantially in understanding its role (and particularly the hypothesized and somewhat novel idea that the amount of glutamate released per vesicle is altered in HSP). They could use sparse knock-down of Rab3A, or simply mix cultures from KO and WT mice (with appropriate tags/labels). The general view in the field has been that HSP is regulated post-synaptically via regulation of AMPAR trafficking, and considerable evidence supports this view. The more support for their suggestion of a pre-synaptic site of control, the better.
2. Rab3A is also found at inhibitory synapses. It would be very informative to know if HSP at inhibitory synapses is similarly affected. This is particularly relevant as at inhibitory synapses, one expects a removal of GABARs and/or a decrease of GABA-packaging in vesicles (ie the opposite of whatever is happening at excitatory synapses). If both processes are regulated by Rab3A, this might suggest a role for this protein more upstream in the signaling; an effect only at excitatory synapses would argue for a more specific role just at these synapses.

Author Response

Thank you for your thorough critique and thoughtful suggestions for improving our manuscript, "Homeostatic Synaptic Plasticity of Miniature Excitatory Postsynaptic Currents in Mouse Cortical Cultures Requires Neuronal Rab3A." The reviewers' detailed comments suggest that showing multiple types of graphs to demonstrate the presence of divergent scaling of mEPSC amplitudes in cultures from Rab3A wild type, and its disruption in cultures from Rab3A knockout mice, had the unintended consequence of obscuring the major results of our study. Furthermore, our proposal that the difference in characteristics of scaling of GluA2 receptor expression compared to that of mEPSC amplitudes, based on the ratio plots, indicated that a mechanism other than postsynaptic receptors likely contributes to the homeostatic increase in mEPSC amplitude was not convincing to the reviewers. Reviewers 2 and 3 point out these results might be explained by differences in the limitations and artifacts of the two very distinct techniques, electrophysiology and fluorescence imaging. In the revision we will acknowledge that a greater variability in the signal, or, more issues with signal over noise, might be present in imaging experiments compared to electrophysiology. This could explain the lack of identical effects on GluA2 receptors compared to mEPSC amplitudes in the matched experiments, but we maintain it is also possible that a greater variability in GluA2 responses is biologically meaningful. Further, an issue with the accuracy of imaging experiments to report the true receptor effects would also call into question the conclusion that receptors always increase after activity blockade. Finally, the graphs illustrating the detailed characteristics of scaling with rank order and ratio plots required pooling multiple samples per cell, which precludes application of standard statistical methods to determine whether effects or differences reach statistical significance. Therefore, we will remove the cumulative distribution functions, rank order plots, and ratio plots, and show only analyses that involve a single sample per cell. This major change will simplify and clarify the main findings, that homeostatic plasticity of both mEPSC amplitude and GluA2 receptor expression in mouse cortical cultures involves the synaptic vesicle protein Rab3A operating in neurons rather than astrocytes. We will focus our comparison between mEPSC amplitudes and receptors in the same cultures to differences between the magnitude of effects on the mean or median, and will make clear that overall, our data can be explained by two possibilities: 1) the presynaptic vesicle protein is acting via regulation of postsynaptic receptors alone, or, it is regulating both postsynaptic receptors and another contributor to mEPSC amplitude, possibly amount of transmitter released by a single vesicle. Either way, it is very surprising that this presynaptic protein is involved in postsynaptic changes, so our results represent a novel contribution to the field of homeostatic plasticity. In sum, the changes we propose should go a long way towards addressing the majority of the reviewers' major critiques.

A related issue raised by the reviewers was that the model describing potential presynaptic mechanisms of Rab3A in homeostatic plasticity was not supported by direct evidence (Figure 10). We meant the model to introduce the possibility of a presynaptic contribution to mEPSC amplitude and to stimulate future research, but clearly did not communicate its speculative nature, neither in the Figure legend nor in our discussion of potential mechanisms. In the revision, we will restrict the model to the direct findings in this study. Additionally, we will state where appropriate, that while previous findings at the mouse NMJ are consistent with a presynaptic role for Rab3A (Wang et al., 2011), in the current study there is no direct evidence for this idea in cortical cultures other than the quantitative differences in the fold increases in mEPSC amplitudes and GluA2 receptors which were assayed in the same cultures.

We will submit a revised version addressing each of the reviewer's concerns and suggestions as described above and below; these major modifications will greatly improve the readability of the manuscript and clarify the main results.

Reviewer #1

Koesters and colleagues investigated the role of the presynaptic small GTPase Rab3A in homeostatic scaling of miniature synaptic transmission in primary mouse cortical cultures using electrophysiology and immunohistochemistry. The major finding is that TTX incubation for 48 hours does not induce an increase in the amplitude of excitatory synaptic miniature events in neuronal cultures derived from Rab3A KO and Rab3A Earlybird mutant mice. NASPM application had comparable effects on mEPSC amplitude in control and after TTX, implying that Ca²⁺-permeable glutamate receptors are unlikely modulated during synaptic scaling. Immunohistochemical analysis revealed an increase in GluA2 puncta size and intensity in wild type, but not Rab3A KO cultures. Finally, they provide evidence that loss of Rab3A in neurons, but not astrocytes, blocks homeostatic scaling. Based on these data, the authors propose a model in which presynaptic Rab3A is required for homeostatic scaling of synaptic transmission through GluA2-dependent and independent mechanisms.

While the title of the manuscript is mostly supported by data of solid quality, many conclusions, as well as the final model, cannot be derived from the results presented. Importantly, the results do not indicate that Rab3A modulates quantal size on both sides of the synapse. Moreover, several analysis approaches seem inappropriate.

The following points should be addressed:

- 1. The model shown in Figure 10 is not supported by the data. The authors neither provide evidence for two different functional states of Rab3A being involved in mEPSC amplitude modulation, nor for a change in glutamate content of vesicles. Furthermore, the data do not fully support the conclusion of a presynaptic role for Rab3A in homeostatic scaling.*

We will revise the model, removing presynaptic mechanisms for Rab3A and restricting it to the direct findings in this study.

- 1. The analysis of mEPSC data using quantile sampling followed by ratio calculation is not meaningful under the tested experimental conditions because of the following reasons:*

(i) The analysis implicitly assumes that all events have been detected. The prominent mEPSC frequency increase after TTX suggests that this is not the case, i.e., many (small) mEPSCs are likely missed under control conditions.

We explicitly addressed the potential contribution of missed mEPSCs that are below threshold in (Hanes et al., 2020). We found that even simulating a threshold of 7 pA, applied to data artificially modified by uniformly multiplying the control data set, did not generate a ratio plot with the increasing ratio over 75% of the data that we observe in the experimental data. Overall, the findings from simulating a threshold and a uniform multiplicative factor illustrate that the threshold issue does not cause major changes to the data. Furthermore, in cultures from Rab3A^{+/+} mice from the Rab3A^{Ebd}/⁺ colony, the mEPSC amplitudes were significantly smaller than those recorded in cultures from Rab3A^{+/+} mice from the Rab3A^{+/+} colony (lines 327-329, 11 pA vs 13 pA), indicating that if there were smaller mEPSCs occurring in the Rab3A^{+/+} data set, we would have detected them. Although for these reasons we feel it is unlikely our ratio plot analysis is invalid, to clarify the result that homeostatic plasticity of mEPSC amplitude requires functioning Rab3A, we will remove the ratio plots.

(ii) The analysis is used to conclude how events of a certain size are altered by TTX treatment. However, this analysis compares the smallest mEPSCs of the TTX condition

with the smallest control mEPSCs, but this is not a pre-post experimental design. Variation between cells and between coverslips will markedly affect the results and lead to misleading interpretations.

The rank order plot is a well-established plot to examine the mathematical transformation caused by homeostatic plasticity, first used in (Turrigiano et al., 1998). We included it here to facilitate comparison of our findings with previous results. We introduced the ratio plot in (Hanes et al., 2020), finding it shows more clearly differences occurring in the range of small mEPSC values. The reviewer is correct in that we are assuming the smallest mEPSCs before treatment should be matched with the smallest mEPSCs after treatment. It is almost impossible to do a pre-post experimental design for mEPSCs. Even when applying a treatment, for example acute perfusion with a receptor antagonist, to a single cell and recording mEPSCs before and after the treatment, it is not a true pre-post design at the level of mEPSC amplitudes, which come from many different inputs. The power of the method is that different characteristic mathematical transformations for different experimental conditions (e.g., genotype or activity protocol) support the idea that those conditions either involve different mechanisms or have altered the mechanism. Such differences might be missed by only comparing means or medians. However, we found no evidence that loss of Rab3A or expression of the Rab3A Earlybird mutant altered the mathematical transformation due to homeostatic plasticity, other than to reduce its magnitude across all amplitudes. Therefore, including these complex analyses is not adding anything to the finding that Rab3A plays a role in homeostatic plasticity of mEPSC amplitudes and they will be removed in the revision.

(iii) The ratio (TTX/control) vs. control plots seem to suffer from a division by small value artifact (see Figure 6F).

The reviewer is referring to findings on the ratio plot for receptor cluster area. Because the large ratios for the smallest control areas occur in the cultures prepared from wild type mice, and to a much lower extent in cultures prepared from Rab3A knockout mice, we think there is a biologically relevant increase in the TTX/CON ratio, since an artifact due to division by small values should be present in both data sets. However, we cannot rule out that the differences in ratio plot behavior between receptors and mEPSC amplitudes result from the different limitations in detection of receptor clusters vs. the limits of detection of mEPSCs, so we will remove the ratio plots and focus on comparison of means or medians.

Correspondingly, ratio-analysis differs considerably for different control conditions (Fig. 1Giii, Fig. 2Giii, Fig. 6C, Fig. 9A).

The reviewer is correct to point out that the ratio plot shows differences across control conditions (note, these differences are not obvious with the more standard rank order plot). The magnitude of the 50th percentile ratio differs across control conditions, and behaviors of the largest mEPSCs also differ, with some ratios going down at the highest control amplitudes (1Giii, 6C), and others continuing to increase with increasing control amplitude (2Giii, 9A). They all share the divergent increasing ratio from smallest mEPSC amplitude to around the 20 pA level. We attribute the differences in magnitude to the differences in experimental conditions: 1Giii is Rab3A^{+/+} from the ^{+/+} colony; 1Giii is Rab3A^{+/+} from the Ebd^{+/+} colony; 6C is a set of Rab3A^{+/+} cultures assayed several years after the set in 1Giii; 9A is a different culture condition altogether, with neurons being plated onto an already formed bed of astrocytes. Effects on the largest mEPSCs are likely attributable to the small number and high variability of amplitudes in this range. Since the variability in the very sensitive ratio plot have taken away from the main findings of homeostatic plasticity being disrupted in the absence of functioning Rab3A in neurons, we will remove the rank-order and the ratio plots from the manuscript.

1. As noted by the authors in a previous publication (Hanes et al. 2020), statistical analysis of CDFs suffers from inflation. In addition, the quantile sampling method chosen violates an important assumption of the K-S test. Indeed, p-values for these comparisons are typically several orders of magnitude smaller. Given that the statistical N most likely corresponds to the number of cultures (see, e.g., <https://doi.org/10.1371/journal.pbio.2005282>), CDF comparisons are not informative and should thus not be used to draw conclusions from the data. The plots can be informative, though.

As the reviewer acknowledges, we were very careful in (Hanes et al., 2020) to state that the p values could not be used to determine significance in the KS test of cumulative distributions for pooled data because the KS test assumes a single sample per cell. We also suggested in that study that the p values could be used in a comparative way for looking at data sets with similar (inflated) n values to say something about bigger or smaller differences. We failed to reiterate those caveats here. In reviewing the article “What is N” by (Lazic et al., 2018) (which we very much appreciate being shown by the reviewer), we agree that in the current study where we are attempting to show how the effect of homeostatic plasticity is or is not altered by loss of Rab3A function, it is imperative that we be able to make conclusions about statistical significance. The pooling approach is essential for having some sense of the mEPSC amplitude distributions, but that is not necessary for looking at the effect of Rab3A. Therefore, we will remove all analyses that involve pooling of multiple mEPSC amplitudes per cell.

1. How does recoding noise and the mEPSC amplitude threshold affect “divergent scaling”?

We addressed this in our 2020 paper (Hanes et al., 2020) where we showed that the experimental homeostatic increase in mEPSC amplitude cannot be simulated with uniform, multiplicative synaptic scaling whether we included or excluded distortion caused by a detection threshold.

1. What is the justification for the line fits of the ratio data/how was the fit range chosen?

We are assuming the reviewer is referring to the line fits for the rank-order data. If so, the fit range is the entire range of the data. This issue will be addressed by the removal of the rank-order plots from the manuscript.

1. TTX application induces a significant increase in mEPSC amplitude in Rab3A^{-/-} mice in two out of three data sets (Figs. 1 and 9). Hence, the major conclusion that Rab3A is required for homeostatic scaling is only partially supported by the data.

Based on the p-values for comparison of means with a Kruskal-Wallis test, we would argue that TTX application does not show a significant increase in mEPSC amplitude in Rab3A^{-/-} neurons (Figure 1 p-value = .318; Figure 9 p-value = .125) when comparing to untreated control mEPSC amplitude means. It is only when we use the KS test and the inflated n’s that we get a barely significant results, p = 0.042. Based on the Lazic article (Lazic et al., 2018), we would now conclude that we cannot use the KS p value in that analysis. We have tried to be clear that the effect of TTX application on mEPSC amplitude in Rab3A^{-/-} neurons is not completely abolished, but rather is dramatically reduced, which we acknowledge in the manuscript (line 279). This issue will be addressed by removal of CDFs from the manuscript.

1. Line 289: A comparison of p-values between conditions does not allow any meaningful conclusions.

Although we feel that comparison of magnitude of effects can be stated in a qualitative way for similar sized pooled data sets with larger or smaller p-values, we agree that statistical significance has no meaning. This issue will be addressed by removing the CDF plots from the manuscript.

1. There is a significant increase in baseline mEPSC amplitude in Rab3AEbd/Ebd (15 pA) vs. Rab3Aebd/+ (11 pA) cultures, but not in Rab3A-/- (13.6 pA) vs. Rab3A+/- (13.9 pA). Although the nature of scaling was different between Rab3AEbd/Ebd vs. Rab3Aebd/+, and Rab3AEbd/Ebd with vs. without TTX, the question arises whether the increase in mEPSC amplitude in Rab3AEbd/Ebd is Rab3A dependent. Could a Rab3A independent mechanism occlude scaling?

We have acknowledged in the manuscript that one explanation for a failure to exhibit homeostatic plasticity in the cultures from Rab3A Earlybird mutant mice is that the already large basal amplitude occludes any further increase (line 366). In the revision we will make sure the occlusion possibility is highlighted, but we will also discuss other proteins that have been implicated in homeostatic plasticity that have caused an increase in mEPSC amplitude and/or AMPA receptors at baseline, for example, Arc/Arg3.1 KO (Shepherd et al., 2006; Beique et al., 2011); Homer KO (Hu et al., 2010) and inhibition of mir-186-5p (Silva et al., 2019).

1. Figure 4: NASPM appears to have a stronger effect on mEPSC frequency in the TTX condition vs. control (-40% vs. 15%). A larger sample size might be necessary to draw definitive conclusions on the contribution of Ca²⁺-permeable AMPARs.

We will acknowledge that Ca²⁺-permeable AMPARs could be contributing to the frequency increase following activity blockade and will also include analyses of frequency throughout the manuscript.

1. The authors discuss previous papers showing changes in VGLUT1 intensity. Was VGLUT1 intensity altered in the stainings presented in the manuscript?

We will perform analyses VGLUT1 intensity and include them in the manuscript.

1. The change in GluA2 area or fluorescence intensity upon TTX treatment in controls is modest. How does the GluA2 integral change?

The changes in GluA2 integrals look exactly like the changes in cluster size and were not included to simplify the results. But with the removal of the CDFs, rank order, and ratio plots, we can easily include integral measurements. What we did not observe was an additive effect with intensity and size such that the effects on integral were of greater magnitude or statistical significance than either alone. We will include the integral plots in the revised manuscript.

1. The quantitative comparison between physiology and microscopy data is problematic. The authors report a mismatch in ratio values between the smallest mEPSC amplitudes and smallest GluA2 receptor cluster sizes (l. 464; Figure 8). Is this comparison affected by the fluorescence intensity threshold?

| *What was the rationale for a threshold of 400 a.u. or 450 a.u.?*

We have acquired AOIs of receptor clusters at multiple threshold levels, and can examine whether the results are altered when using a low, medium or high threshold level.

| *How does this threshold compare to the mEPSC threshold of 3 pA?*

The issue with values being below threshold in untreated cultures has been the concern in interpreting effects on mEPSC amplitudes, specifically, whether this mismatch contributes to divergent scaling. A problem of values being below a toohighly set threshold in the control and becoming detectable after the homeostatic plasticity produces a lower ratio than expected, because now there are values in the treated condition that were not present in the control condition. Instead, for GluA2 receptor cluster size, we observed higher TTX/CON ratios at the low end of the data set. So, based on this, the thresholds chosen for imaging are not having the same effect, if that is what is being asked. This issue will be addressed by removing ratio plots.

| *The conclusion that an increase in AMPAR levels is not fully responsible for the observed mEPSC increase is mainly based on the rank-order analysis of GluA2 intensity, yielding a slope of ~0.9. There are several points to consider here: (i) GluA2 fluorescence intensity did increase on average, as did GluA2 cluster size. (ii) The increase in GluA2 cluster size is very similar to the increase in mEPSC amplitude (each approx. 18-20%). (iii) Are there any reports that fluorescence intensity values are linearly reporting mEPSC amplitudes (in this system)?*

We agree that our data show GluA2 receptors increase as based on cluster size, and did not mean to imply otherwise. Our conclusion that there is another contributor to mEPSC amplitude other than receptors is based on two main findings, 1) that the ratio plots for mEPSC amplitudes and receptor cluster size have distinctively different behaviors, and 2) that there are differences in either magnitude or direction of the TTX effect across 6 matched cultures, 3 from WT animals and 3 from TTX animals (see more explanation of this point below, in response to Reviewer 3). To our knowledge, no one has reported homeostatic plasticity effects on a culture by culture basis, and no one has compared imaging results and physiological results for the same cultures. We will remove the ratio plots and the conclusions based on the differences in behavior for mEPSC amplitudes and receptor cluster size. We will acknowledge in the revision that the differences in magnitude and direction across the 6 matched cultures could be due to the differences in limitations and artifacts of imaging fluorescent antibody staining vs. the limitations and artifacts of detecting mEPSCs electrophysiologically. However, we will continue to state that our results could also be due to the possibility that mEPSC amplitude is not changing in lockstep with receptor levels in every situation. To support this proposal, we will discuss those articles that include both measurements, and point out where mEPSC amplitude measurements and receptor levels match and where they do not.

| *Antibody labelling efficiency, and false negatives of mEPSC recordings may influence the results. The latter was already noted by the authors.*

We will add the caveat that antibody labeling efficiency can vary between coverslips. Although we prepared single solutions that were applied to all coverslips in an experiment, this was not possible for the primary antibody to GluA2, which was added to live cultures in individual wells.(iv) It is not entirely clear if their imaging experiments will sample from all synapses. We will add to Materials and Methods that we sample from all the synapses that could be detected by the researcher on the primary dendrite of the pyramidal cell.

Other AMPAR subtypes than GluA2 could contribute, as could kainate or NMDA receptors.

This is true, other AMPARs (GluA3 and/or GluA4) could be contributing, but we only looked at the receptors well established to be contributing to homeostatic plasticity (GluA1 and GluA2). We will acknowledge the possible contribution of other AMPARs in the revised manuscript.

Furthermore, the statement "complete lack of correspondence of TTX/CON ratios" is not supported by the data presented (l. 515ff). First, under the assumption that no scaling occurs in Rab3A^{-/-}, the TTX/CON ratios show a 20-30% change, which indicates the variation of this readout. Second, the two examples shown in Figure 8 for Rab3A^{+/+} are actually quite similar (culture #1 and #2), particularly when ignoring the leftmost section of the data, which is heavily affected by the raw values approaching zero.

We will remove the ratio plots from the manuscript and the arguments about differences between GluA2 receptors and mEPSC amplitudes that were based on them. However, we maintain that we have demonstrated a lack of consistent effect for GluA2 receptors and mEPSCs in the matched culture experiments. Yes, the readout of homeostatic plasticity in ratio plots for mEPSCs in the Rab3AKO reach over 1.1 in Figure 1, and as high as 1.2 in the cultures where Rab3AKO neurons were plated on Rab3AWT glia (Figure 9). Our point is that if we had measured GluA2 receptor responses to TTX in those same experiments, the ratios should have been above 1. However, in the experiments in which we measured both mEPSCs and GluA2 receptors, the ratios do not match. In culture #1, the ratio for mEPSCs was at 1 for more than 50% of the data, but for GluA2 receptors, was below 1 for more than 50% of the data. In culture #3, the ratio for mEPSCs was below 1 for more than 50% of the data, but for GluA2 receptors was close to 1.2 for 50% of the data. Only for culture #2 do the ratios appear to match. In the revised manuscript, the evidence that GluA2 receptors and mEPSCs are not changing in parallel will be based on the behavior of means or medians in untreated vs TTX-treated cultures, rather than ratio plots. It could be argued that we need a greater number of matched experiments to make conclusions, but the whole point of a matched experiment is that it should always show the same result—we are no longer dealing with the variability in the homeostatic plasticity itself. We will add a statement that the only three explanations left for the failure of mEPSC amplitudes and GluA2 receptors to change in parallel are 1) a true mismatch, 2) a sampling issue, or 3) technical artifacts that occur in one culture and not another.

1. Figure 7A: TTX CDF was shifted to smaller mEPSC amplitude values in Rab3A^{-/-} cultures. How can this be explained?

Figure 7A depicts the pooled data that are shown separately for 3 cultures in Figure 8. We observed mEPSC amplitudes being smaller after TTX treatment in some range of the data for all three Rab3AKO cultures, suggesting that this may be a biological result rather than random variation around no change (which would be a ratio of 1). However, this effect is not significant at the level of means, nor in the KS test (which has the issue of inflated n in any case), so we did not highlight this point. This issue will be addressed by the removal of the CDF plots from the manuscript.

Reviewer #2

Technical concerns:

1. The culture condition is questionable. The authors saw no NMDAR current present during spontaneous recordings, which is worrisome since NMDARs should be active in cultures with normal network activity (Watt et al., 2000; Sutton et al., 2006).

The (Watt et al., 2000) study recorded mEPSCs in 0 Mg²⁺ (Figure 1). The (Sutton et al., 2006) study also shows an average mEPSC waveform (Figure 1D) that was recorded from in 0 Mg²⁺. Our extracellular recording solution contains Mg²⁺ (1.3 mM) so we likely are not observing NMDA-mediated currents because they are blocked with Mg²⁺ when strong depolarizations are prevented with TTX in the recording solution. We will add the idea that the NMDA currents are blocked by Mg²⁺ to Material and Methods.

It is important to ensure there is enough spiking activity before doing any activity manipulation.

We agree that it would be best if network spiking activity were monitored alongside mEPSC recordings, for example by culturing on multi-electrode arrays. Data from these measurements might explain culture to culture variability in homeostatic responses. To our knowledge, most other studies investigating homeostatic plasticity do not monitor network spiking activity in the same cultures that assay mEPSC amplitudes. This is something that the field should move towards. We will add the caveat that activity was not directly measured to the manuscript.

Similarly, it is also unknown whether spiking activity is normal in Rab3A KO/Ebd neurons.

Since we did not measure spiking activity, we cannot address whether the disruption in homeostatic plasticity in cultures prepared from Rab3A KO and Rab3A^{Ebd}/Ebd mutant mice is due to an alteration in network activity. If activity were already low in cultures prepared from these genetically altered mice, we would expect mEPSC amplitudes to be increased, compared to those measured in cultures from WT animals. That is not the case in cultures from Rab3A KO mice, so it is unlikely that network activity is reduced. However, mEPSC amplitudes are increased in Rab3A^{Ebd}/Ebd cultures, leaving open this possibility. It would have to be a defect unique to neurons in culture, since the Rab3A^{Ebd}/Ebd mouse appears normal in every way, suggesting action potential activity is occurring in the brains of these animals in vivo. We will add the possibility that activity is altered in the cultures from Rab3A KO and Rab3A^{Ebd}/Ebd to the manuscript.

1. Selection of mEPSC events is not conducted in an unbiased manner. Manually selecting events is insufficient for cumulative distribution analysis, where small biases could skew the entire distribution. Since the authors claim their ratio plot is a better method to detect the uniformity of scaling than the well-established rank-order plot, it is important to use an unbiased population to substantiate this claim.

MiniAnalysis (a standard program used for mEPSC event detection and analysis) selects many false positives with the automated feature (due to the very small sizes of events that are close to the noise level) so manual re-evaluation of the automated process is necessary to eliminate false positives. As soon as there is a manual step, bias is introduced. Interestingly, a manual reevaluation step was applied in a recent study that describes their process as ‘unbiased’ (Wu et al., 2020). The alternative is to apply a very large threshold, reducing or eliminating false positives. However, this has the effect of biasing the data towards large events. In sum, we do not believe it is currently possible to perform a completely unbiased detection process. We feel that it is important to include as many small events as possible to reduce the problem of having events in the TTX experimental group that were not matched by events in the

control experimental group, for the rank order and ratio plots, so setting the threshold low and manually detecting events accomplishes this. We will add to the Materials and Methods section that the person selecting events did not have information on whether the record was from an untreated or a TTX-treated cell at the time of selection. All of these issues, the potential for skewing the CDFs, and bias potentially interfering in the true rank order and ratio relationships, are addressed by removal of the CDFs, ratio and rank-order plots from the manuscript.

1. Immunohistochemistry data analysis is problematic. The authors only labeled dendrites without doing cell-fills to look at morphology, so it is questionable how they differentiate branches from pyramidal neurons and interneurons. Since glutamatergic synapses on these two types of neuron scale in the opposite directions, it is crucial to show that only pyramidal neurons are included for analysis.

MAP2, in addition to labeling dendrites, also labels the cell body, and we used the cell structure revealed by MAP2 staining to select pyramidal-shaped neurons. The selection of the primary dendrite of a pyramidal neuron was stated in lines 239-240 in Materials and Methods and lines 1094 in the figure legend, but we had not explicitly stated how we knew it was a pyramidal neuron. We will include a low power picture of each of the selected pyramidal neurons in the revision.

Conceptual concerns:

The only novel finding here is the implicated role for Rab3A in synaptic scaling, but insights into mechanisms behind this observation are lacking. The author claims that Rab3A likely regulates scaling from the presynaptic side, yet there is no direct evidence from data presented. In its current form, this study's contribution to the field is very limited.

We acknowledge that a presynaptic mechanism is involved in the regulation of homeostatic plasticity by Rab3A is not supported by direct evidence in cortical cultures in this study. But we disagree that the study's contribution is very limited.

The revised manuscript will emphasize that there are only two possible mechanisms by which Rab3A is acting in homeostatic plasticity. Either this presynaptic vesicle protein is regulating postsynaptic receptors (an extremely surprising result for which we do have direct evidence), or, it is regulating quantal size from both sides of the synapse (supported by direct evidence from our previous study at the mouse neuromuscular junction in vivo, where receptors are not being upregulated during homeostatic plasticity, and, by indirect evidence in the current study, that receptors and mEPSCs are not being identically regulated in the same cultures). Furthermore, the first idea that follows from the effect of Rab3A on receptors is that it would be regulating release of factors from astrocytes, since this is a mechanism that has been shown to be involved in homeostatic plasticity, and we clearly disprove this hypothesis.

1. Their major argument for this is that homeostatic effects on mEPSC amplitudes and GluA2 cluster sizes do not match. This is inconsistent with reports from multiple labs showing that upscaling of mEPSC amplitude and GluA2 accumulation occur side by side during scaling (Ibata et al., 2008; Pozo et al., 2012; Tan et al., 2015; Silva et al., 2019).

We agree with the reviewer that many studies show an increase in receptors and mEPSC amplitudes after activity blockade. This is why we were very surprised in our initial experiments to find that there was not a consistent robust increase in receptors in our cultures. At that point we were only imaging, and we assumed that it was homeostatic

plasticity that was not always robust. We decided it was essential to measure mEPSC amplitudes and image receptors in the same cultures. We expected to observe larger and smaller effects on mEPSC amplitudes from culture to culture that were paralleled by larger and smaller effects on receptors, but this is not what happened. We have gone back to the literature to look more closely at whether variability across cultures has ever been shown for mEPSC amplitudes, receptors, or both. In a survey of 14 studies, none report results culture by culture. To our knowledge, we are the first to report this variability in the receptor response, and the lack of correlation between mEPSC amplitudes and receptor responses, in the same cultures. That said, for the 4 examples provided by the reviewer, only 1 reports evidence relevant to our study that receptors and mEPSC amplitudes ‘occur side by side,’ which is the (Ibata et al., 2008) study. Here, 24 hr of TTX treatment of rat cortical cultures causes synaptically localized GluA2 receptors in confocal imaging, and mEPSC amplitudes, to both increase to around 130%. The (Pozo et al., 2012) study is not a study of activity blockade but of the effects of overexpressing beta-integrins in rat hippocampal cultures, and this causes both GluA2 receptors and mEPSC amplitudes to increase, but the GluA2 level is not restricted to synaptic sites, and, is expressed as the surface fraction (surface receptor/total receptor—total receptor being surface intensity plus internalized intensity) which increases from 0.5 to 0.55, or to 110%, while mEPSC amplitude increases to ~180%. The (Tan et al., 2015) study only provides Western blot data to show an increase of receptors to 125% in mouse cortical cultures in response to 48 hr TTX, with mEPSC amplitudes increased to ~140%, but the Western blot technique measures synaptic and nonsynaptic receptors on excitatory and inhibitory neurons, as well as receptors on astrocytes. Finally, in (Silva et al., 2019), the culture conditions for the imaging data and the mEPSC amplitude data are markedly different, with ‘low-density’ Banker cultures being used for the former, and ‘high-density’ cultures used for the latter, and the protocol to induce activity blockade is different from ours (noncompetitive AMPA and NMDA blockers); synaptic GluA2 receptors are increased to ~280% and mEPSC amplitudes to ~170%. In the revision we will carefully summarize the previous evidence for receptors and mEPSC amplitude responses to activity blockade. Since it is known that different protocols trigger different molecular mechanisms, for example, TTX + APV triggers a homeostatic plasticity that can be completely reversed by acute application of blockers of Ca-permeable receptors, whereas TTX alone triggers a plasticity that is insensitive to these blockers (Sutton et al., 2006), Figure 4E; (Soden and Chen, 2010); Figure 4A), we will keep our discussion restricted to studies using TTX alone for at least 24 hr. We will acknowledge that our finding that GluA2 receptors and mEPSC amplitudes are not varying in lockstep from culture to culture suggests there is another contributor to mEPSC amplitude, but that we cannot rule out it is due to a greater variability in signal, or more issues with signal over noise, in imaging experiments compared to electrophysiology experiments.

Studies surveyed about reporting results by culture:

(Ju et al., 2004; Stellwagen et al., 2005; Shepherd et al., 2006; Sutton et al., 2006; Cingolani and Goda, 2008; Hou et al., 2008; Ibata et al., 2008; Chang et al., 2010; Hu et al., 2010; Jakawich et al., 2010; Beique et al., 2011; Tatavarty et al., 2013; Diering et al., 2014; Sanderson et al., 2018)

Further, because the acquisition and quantification methods for mEPSC recordings and immunohistochemistry imaging are entirely different (each with its own limitations in signal detection), it is not convincing that the lack of proportional changes must signify a presynaptic component.

We agree with the reviewer that there is no way to compare absolute levels from one type of experimental technique to another, but whatever differences in technical issues there are for the two techniques, they should cause systemic errors and should not contribute to the differences between experiments. Most of the issues with imaging come down to variability in the intensity of fluorescence from experiment to experiment, since the antibody solutions are made anew each time, as is the fixation solution. In addition, the confocal microscope

function can vary over time and give brighter or dimmer images. But those kinds of artifacts are addressed by using the same solutions on control and TTX-treated coverslips, and imaging control and TTX-treated coverslips in the same single 2-3 hour imaging session, so that whatever issues there are, they cannot contribute to the TTX effect itself. Therefore when we compare the TTX effect (TTX measurements compared to untreated measurements) from culture to culture and find that in one WT culture there was no increase in receptors but there was in mEPSC amplitude, it is difficult to explain how a limitation specific to the antibody imaging technique could produce such a result. Similarly, when we get the opposite result, that in one KO culture, receptors increased but mEPSC amplitudes did not, it is unclear how limitations in signal detection would produce such a result in one culture but not another. The one exception to this is that the primary GluA2 antibody has to be added individually to each coverslip before returning the dishes to the incubator in order to avoid the disruption to live cells that a complete removal of media would have had. The only remaining ‘artifact’ that could explain the results would be a greater variability in the imaging experiments due to limitations in the signal or the signal to noise ratio. In the revision we will report additional characteristics of imaging experiments, such as average intensity for each coverslip, and for each experiment, to address whether variability in fluorescence levels could explain the variability in TTX effects we observe. We will include the possibility that the mismatches in GluA2 receptors and mEPSCs could be caused by greater variability in the imaging experiments.

1. The authors also speculate in the discussion that presynaptic Rab3A could be interacting with retrograde BDNF signaling to regulate postsynaptic AMPARs. Without data showing Rab3A-dependent presynaptic changes after TTX treatment, this argument is not compelling. In this retrograde pathway, BDNF is synthesized in and released from dendrites (Jakawich et al., 2010; Thapliyal et al., 2022), and it is entirely possible for postsynaptic Rab3A to interfere with this process cell-autonomously.

In the revision, the model will focus on the direct findings of the manuscript and tone down the speculation about BDNF signaling, but in the Discussion we will add the possibility that a Rab3A-BDNF interaction could occur either presynaptically or postsynaptically. Interestingly, these articles suggest the postsynaptic BDNF is affecting presynaptic function, namely mEPSC frequency. It is conceivable it could presynaptically affect the vesicle’s release of transmitter.

1. The authors propose that a change in AMPAR subunit composition from GluA2-containing ones to GluA1 homomers may account for the distinct changes in mEPSC amplitudes and GluA2 clusters. However, their data from the NASPM wash-in experiments clearly show that GluA1 homomer contributions have not changed before and after TTX treatment.

Our apologies to the reviewer that we were not clear on this point. In lines 396 to 400 we were describing the significant effects that NASPM had on mEPSC frequency on both untreated and TTX-treated cells, despite having only modest, and not quite significant effects on mEPSC amplitude. We conclude from these results that there are synaptic sites that have only GluA1 homomers, and the mEPSCs from these sites are blocked 100% by NASPM. There may be an increase in such GluA1-only synapses after activity blockade, but nevertheless, these events do not contribute to the amplitude increase. So we did not mean to suggest that there is a shift from GluA2 containing to GluA1 containing receptors that leads to the amplitude increase and fully agree with the reviewer that the GluA1 homomer contributions to amplitude have not changed before and after TTX. We will clarify the difference between the contribution of GluA1 homomers to amplitude and frequency in the revised manuscript.

Reviewer #3

Summary: The authors clearly demonstrate the Rab3A plays a role in HSP at excitatory synapses, with substantially less plasticity occurring in the Rab3A KO neurons. There is also no apparent HSP in the Earlybird Rab3A mutation, although baseline synaptic strength seems already elevated. In this context, it is unclear if the plasticity is absent or just occluded by a ceiling effect due the synapses already being strengthened. The authors do appropriately discuss both options. There are also differences in genetic background between the Rab3A KO and Earlybird mutants that could also impact the results, which are also noted. The authors have solid data showing that Rab3A is unlikely to be active in astrocytes. Finally, they attempt to study the linkage between synaptic strength during HSP and AMPA receptor trafficking, and conclude that trafficking is largely not responsible for the changes in synaptic strength.

Strengths: This work adds another player into the mechanisms underlying an important form of synaptic plasticity. The plasticity is only reduced, suggesting Rab3A is only partially required and perhaps multiple mechanisms contribute. The authors speculate about some possible novel mechanisms.

Weaknesses: However, the rather strong conclusions on the dissociation of AMPAR trafficking and synaptic response are made from somewhat weaker data. The key issue is the GluA2 immunostaining in comparison with the mEPSC recordings. Their imaging method involves only assessing puncta clearly associated with a MAP2 labeled dendrite. This is a small subset of synapses, judging from the sample micrographs (Fig 5). To my knowledge, this is a new and unvalidated approach that could represent a particular subset of synapses not representative of the synapses contributing to the mEPSC change. (they are also sampling different neurons for the two measurements; an additional unknown detail is how far from the cell body were the analyzed dendrites for immunostaining. While the authors acknowledge that a sampling issue could explain the data, they still use this data to draw strong conclusions about the lack of AMPAR trafficking contribution to the mEPSC amplitude change. This apparent difference may be a methodological issue rather than a biological one, and at this point it is impossible to differentiate these. It will unfortunately be difficult to validate their approach. Perhaps if they were to drive NMDA-dependent LTD or chemLTP, and show alignment of the imaging and ephys, that would help. More helpful would be recordings and imaging from the same neurons but this is challenging. Sampling from identified synapses would of course be ideal, perhaps from 2P uncaging combined with SEP-labeled AMPARs, but this is more challenging still. But without data to validate the method, it seems unwarranted to make such strong conclusions such as that AMPAR trafficking does not underlie the increase in mEPSC amplitude, given the previous data supporting such a model.

We chose the primary dendrite to ensure we were not assaying dendrites from inhibitory neurons or on axons, but we will add in the revision that it is a limitation of our methods that we are not sampling all the synapses for each neuron. The majority of previous studies that establish that receptors are increased side by side with mEPSCs did not measure receptors and mEPSCs in the same cells, nor even in the same cultures. There is a recent study which employs dual recordings, transfection of GluA2 and VGlut1 constructs, and infusion of dyes to highlight cell morphology (Letellier et al., 2019), so in principle an experiment could be done in which synaptic GluA2 sites are imaged in a cell in which the mEPSCs are also measured. It would be difficult to make these measurements in the same cells before and after TTX treatment, since there is a high likelihood of damaging the cell upon electrode withdrawal and with the imaging process itself. In theory, only a few such experiments would be necessary to establish whether receptors and mEPSC amplitudes are varying in lockstep, and we will consider this for a future study. As stated in response to conceptual concern #1 in

Reviewer 2's comments, we will review the literature on previous studies' demonstrations of increases in receptors and mEPSC amplitudes following activity blockade in more detail, including how the synaptic sites to be imaged were chosen, to address whether our selection of sites touching the primary dendrite is unvalidated.

A sample from 3 articles:

(Ibata et al., 2008), only information is that 'distal dendrites' were examined. The authors do not use a dendritic label. (Jakawich et al., 2010), 'neurons with pyramidal-like morphology were selected for imaging,' and 'principal dendrite of each neuron was linearized'—but how these were identified is not clear, since MAP2 or other cellular labels are not described.

(Silva et al., 2019), 'dendrites with similar thickness and appearance were randomly selected using MAP2 staining,' which suggests synaptic sites with GluA2 and VGLUT1 were selected on the basis of being close to or touching the MAP2 positive dendrite, although this is not stated explicitly.

We can perform length measurements on the dendrites imaged and report this information in the revision, but the primary dendrite is the closest dendrite to the cell body.

We have addressed the potential contribution of technical artifacts arising from the two distinct methods of measurement, imaging and electrophysiology, in our response to conceptual concern #1 of Reviewer 2.

Other questions arise from the NASPM experiments, used to justify looking at GluA2 (and not GluA1) in the immunostaining. First, there is a frequency effect that is quite unclear in origin. One would expect NASPM to merely block some fraction of the post-synaptic current, and not affect pre-synaptic release or block whole synapses. It is also unclear why the authors argue this proves that the NASPM was at an effective concentration (lines 399-400).

We observed a clear effect of NASPM reducing mEPSC frequency. We will state more clearly that we infer from the loss of mEPSCs after NASPM that such mEPSCs were from synaptic sites that had only GluA1 homomers, and acknowledge that this is an interpretation. We will also clarify that if our inference is correct, it would indicate that the dose of NASPM we used was 100% effective at blocking GluA1 homomers. The alternative explanation would be a presynaptic effect of NASPM, which has never been reported, to our knowledge.

Further, the amplitude data show a strong trend towards smaller amplitude. The p value for both control and TTX neurons was 0.08 - it is very difficult to argue that there is no effect. And the decrease is larger in the TTX neurons. Considering the strong claims for a pre-synaptic and the use of this data to justify only looking at GluA2 by immunostaining, these data do not offer much support of the conclusions. Between the sampling issues and perhaps looking at the wrong GluA subunit, it seems premature to argue that trafficking is not a contributor to the mEPSC amplitude change, especially given the substantial support for that hypothesis. Further, even if trafficking is not the major contributor, there could be shifts in conductance (perhaps due to regulation of auxiliary subunits) that does not necessitate a pre-synaptic locus. While the authors are free to hypothesize such a mechanism, it would be prudent to acknowledge other options and explanations.

We did not mean to suggest that there is no effect of NASPM on mEPSC amplitude. We will clarify that our data indicate that there is no effect of NASPM on the TTX effect on mEPSC amplitude. We agree with the reviewer that the effect of NASPM on frequency is of larger magnitude after TTX treatment, although the p value is larger than that for untreated cells, likely due to greater variability. We interpret this to mean that TTX treatment increases the

proportion of synapses that have only GluA1 homomers. Nevertheless, the increase in GluA1 homomer sites does not appear to contribute to the overall increase in amplitude following TTX treatment, and we wanted to find the mechanism of the amplitude increase. That is why we focused on GluA2 receptors. We will acknowledge the limitation of basing our conclusions on only GluA2 receptors in the revision, as well as the possibility that there is a change in conductance. As stated in our response to Reviewer 2, we do not mean to state that GluA2 receptors do not go up after activity blockade, we find that this is the case. We are proposing an additional mechanism contributing to mEPSC amplitude to explain the different responses for GluA2 receptors vs. mEPSC amplitudes in some of the 6 matched experiments (3 WT and 3 KO).

The frequency data are missing from the paper, with the exception of the NASPM dataset. The mEPSC frequencies should be reported for all experiments, particularly given that Rab3A is generally viewed as a pre-synaptic protein regulating release. Also, in the NASPM experiments, the average frequency is much higher in the TTX treated cultures. Is this statistically above control values?

We will report frequency measurements for all experiments shown. Following TTX treatment, frequency variability increases enormously, with cells having as high as > 10 mEPSCs per second, and other TTX-treated cells with frequencies as low as < 1 mEPSC per second, so the TTX effect on frequency, and whether this effect is present or not in Rab3A KO and Rab3AEbd/Ebd is not completely clear, which is why we did not include those results previously.

Unaddressed issues that would greatly increase the impact of the paper:

1. Is Rab3A acting pre-synaptically, post-synaptically or both? The authors provide good evidence that Rab3A is acting within neurons and not astrocytes. But where it is acting (pre or post) would aid substantially in understanding its role (and particularly the hypothesized and somewhat novel idea that the amount of glutamate released per vesicle is altered in HSP). They could use sparse knock-down of Rab3A, or simply mix cultures from KO and WT mice (with appropriate tags/labels). The general view in the field has been that HSP is regulated post-synaptically via regulation of AMPAR trafficking, and considerable evidence supports this view. The more support for their suggestion of a pre-synaptic site of control, the better.

We agree with the reviewer that this is the most important question to answer next. The approach suggested by the reviewer would be to record from Rab3A KO neurons in a culture where the majority of its inputs are Rab3A positive. If the TTX effect is absent from these cells, it would strongly indicate that postsynaptic Rab3A is required for homeostatic plasticity. There are not currently transgenic mice expressing GFP forms of Rab3A, so we would have to create one, or, transiently transfect Rab3A-GFP into Rab3AKO neurons. Given that under our experimental conditions, we require a very high density of neurons to observe the increase in mEPSC amplitude, it would be difficult to get the ratio of Rab3A-expressing neurons high enough using transfection to be sure that a given postsynaptic cell lacking Rab3A had a normal number of Rab3A-positive inputs and almost no Rab3A-negative inputs. It may be that the opposite experiment is more doable—an isolated Rab3A-positive neuron in a sea of Rab3A-negative neurons, which could be accomplished with a very low transfection efficiency. Another approach would be to use the fast off rate antagonist gamma-DGG, which is more effective against low glutamate concentrations than high glutamate concentrations (see (Liu et al., 1999; Wu et al., 2007)). If gamma-DGG were less effective at reducing mEPSC amplitude in TTX-treated cells, compared to untreated cells, it would support the hypothesis that activity blockade leads to an increase in the amount of transmitter per vesicle fusion event. Further, if the change in gamma-DGG sensitivity after activity blockade were disrupted

in cultures from Rab3A KO cells, it would support a presynaptic role for Rab3A in homeostatic plasticity of mEPSC amplitude. We have begun these experiments but are finding the surprising result that within a single recording, small mEPSCs and large mEPSCs appear to be differentially sensitive to gamma-DGG. To confirm that this is a biological characteristic, rather than an issue with the detection threshold, we will be repeating our experiments with a slow off rate antagonist that has same effect regardless of transmitter concentration. The complexity of these results precludes including them in the current manuscript.

1. Rab3A is also found at inhibitory synapses. It would be very informative to know if HSP at inhibitory synapses is similarly affected. This is particularly relevant as at inhibitory synapses, one expects a removal of GABARs and/or a decrease of GABA-packaging in vesicles (ie the opposite of whatever is happening at excitatory synapses). If both processes are regulated by Rab3A, this might suggest a role for this protein more upstream in the signaling; an effect only at excitatory synapses would argue for a more specific role just at these synapses.

The next question, after it is determined where Rab3A is acting, is whether it is required for other forms of homeostatic plasticity. This includes plasticity of GABA mIPSCs on pyramidal neurons, but also mEPSCs on inhibitory neurons, and, the downscaling of mEPSCs (and upscaling of mIPSCs) when activity is increased, by bicuculline for example. We will add a statement about future experiments examining other forms of plasticity to the discussion, and include examples where a molecular mechanism has mediated multiple forms, and those that have been shown to be very specific.

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